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Research Paper

GCR: A Global Cable Routing Planner for Two-Dimensional Cluttered Environments

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Abstract

Routing a deformable linear object (DLO), such as a cable or a rope, through a cluttered environment requires more than a collision-free goal shape: every intermediate deformation of the object must also avoid the obstacles. This paper presents the Global Cable Routing (GCR) planner, which computes a sequence of planar cable configurations connecting a given start shape to a given goal shape. The cable is modeled as an ordered polygonal chain, and the search operates directly in the resulting configuration space. Candidate configurations are generated by a guided sampler, shifted by a configuration-level artificial potential field toward the opposite tree of a bidirectional search and away from obstacles, and corrected by a projected elastic relaxation that limits changes in segment vectors and discrete curvature relative to a reference configuration. A tree edge is accepted only if the swept envelope of the local deformation is collision-free and the endpoint correspondence of the ordered chain is preserved; the same test governs parent selection, tree connection, and shortcut refinement. In ten benchmark scenarios, GCR succeeded in all 1000 trials per scenario, with average planning times between 0.008 s and 1.31 s and refined paths of 3–9 configurations. In a dual-arm MuJoCo simulation, tracking the planned endpoint trajectories reproduced the goal shapes with root-mean-square errors of 1.6–4.7 mm, below 2% of the cable length.

Keywords: deformable linear objects, cable routing, motion planning, sampling-based planning, artificial potential fields, dual-arm manipulation.

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1. Introduction

Deformable linear objects (DLOs), such as cables, ropes, wires, hoses, and sutures, appear in different robotic applications, including wire-harness assembly, cable routing, medical suturing, and domestic manipulation. Despite their practical importance, robotic manipulation of DLOs remains difficult because the object shape is not described by a small set of rigid-body coordinates. Instead, the object may bend, stretch slightly, slide around obstacles, and form many admissible geometries under the same endpoint positions. Planning and control for DLOs must therefore handle the object geometry, its deformation behavior, and collision avoidance together, while keeping the commanded shapes mechanically plausible (Sanchez et al. 2018; Yin et al. 2021; Zhu et al. 2022; Caporali and Palli 2026).

The majority of the prior work on DLO manipulation addresses local shape control, where the robot drives the object from its current shape toward a desired one using visual feedback, model-based prediction, or an estimated deformation Jacobian; our earlier studies on dual-arm cable shape control (Almaghout and Klimchik 2022*a*,b, 2024; Almaghout et al. 2024) belong to this line. These methods assume that the target shape, or a sequence of intermediate profiles, is already available. The question of how the cable should travel across a cluttered workspace—through which intermediate shapes, and along which collision-free deformations—is left open.

In contrast, global cable routing requires a different planning formulation. Two cable shapes may be individually collision-free while the swept region between them intersects an obstacle, so collision checking of isolated configurations is insufficient. The planning problem must instead be defined in the configuration space of the complete cable, and tree edges must be validated as local deformations rather than treated only as connections between nearby states.

Classical sampling-based motion planning methods, such as RRT, RRT-Connect, and RRT*, are attractive for high-dimensional problems because they avoid explicit construction of the complete configuration-space obstacle region (Kuffner and LaValle 2000; Karaman and Frazzoli 2011). However, direct application of these methods to DLOs is inefficient because random polygonal chains are usually mechanically invalid, excessively stretched, or collision-prone. Earlier DLO planners addressed this problem by sampling stable or minimum-energy shapes (Moll and Kavraki 2006), by characterizing equilibrium manifolds of elastic rods (Bretl and McCarthy 2014), or by planning elastic-rod motion with contacts (Roussel et al. 2020). These approaches describe rod mechanics rigorously, but high-fidelity rod models and contact-rich planning can be computationally expensive for online routing in cluttered workspaces.

This paper proposes the Global Cable Routing (GCR) planner for a cable-like DLO in a two-dimensional cluttered workspace.

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The cable is represented as an ordered sequence of points connected by fixed-length segments, and the search takes place in the resulting discretized configuration space. Local deformation between neighboring configurations is measured through segment-vector and discrete-curvature changes, which lets the planner penalize unnecessary stretching, compression, rotation, and bending.

The proposed planner couples global tree search with local shape correction. A bidirectional pair of trees preserves global exploration. Each raw sample, drawn by a guided mixture sampler, is refined in two stages: an artificial potential field (APF) acting on the whole configuration pulls it toward the opposite tree and pushes its nodes off nearby obstacles, and a projected elastic relaxation then repairs the biased shape so that its segment lengths and bending remain close to those of a reference configuration. This reference-based regularization keeps the deformation local with respect to the previously feasible configuration instead of driving the cable toward a globally low-energy shape. A central part of the formulation is the ordered transition predicate. Since the cable is represented as an ordered sequence of points, reversing only one of two neighboring configurations changes the point-to-point correspondence used during deformation, and hence the swept region and the collision result. GCR therefore validates each accepted edge with an ordered edge-feasibility predicate that combines swept-envelope collision checking, endpoint-axis consistency, and optional robot-dependent endpoint feasibility, and it applies this predicate at every step where an edge enters a tree or the final path, not only after a path has been generated. The main contributions of this paper are as follows:

1. A formulation of planar cable routing as a search in the configuration space of an ordered polygonal chain, in which both the admissibility of a configuration and the feasibility of a transition are defined relative to a nearby reference configuration.
2. A bidirectional sampling-based planner that generates candidate configurations by guided sampling, biases them with a configuration-level APF, and corrects them by projected elastic relaxation.
3. An ordered transition predicate that combines swept-envelope collision checking with endpoint-correspondence constraints and is applied during parent selection, tree connection, and shortcut refinement.
4. An evaluation of the planner on ten planar benchmark scenarios, together with a dual-arm simulation study in which the planned configuration sequences are executed through endpoint tracking.

The remainder of the paper is organized as follows. Section 2 reviews related work on local shape control, physics-based DLO planning, combined planning–control frameworks, and task-specific cable routing. Section 3 formulates the cable-routing problem. Section 4 presents the proposed GCR algorithm. Section 5 reports the benchmark and robotic validation results. Section 6 discusses the results, limitations, and future directions. Section 7 concludes the paper.

2. Related Work

Research on robotic manipulation of deformable linear objects spans shape control, routing, insertion, transport, and manipulation in constrained environments (Sanchez et al. 2018; Yin et al. 2021; Zhu et al. 2022; Caporali and Palli 2026). Within this field, global planning for cable-like objects in cluttered environments remains less explored than local shape control or task-specific manipulation. This section reviews the four directions most relevant to the present work: local shape control, physics-based DLO planning, combined planning–control frameworks, and task-specific cable routing.

Local shape-control methods assume that a desired shape or a sequence of desired shapes is already given, and the controller computes robot motions to reduce the shape error. Jacobian-based methods estimate a local relation between robot motion and DLO deformation, while learning-based methods approximate this relation from data (Navarro-Alarcon et al. 2013; Jin et al. 2019; Lagneau et al. 2020; Lv et al. 2021).

Our earlier work followed this local shape-control direction. In Almaghout and Klimchik (2022*a*), a planar cable was represented as a set of points, and an analytical Jacobian was derived to map the motion of two robot end-effectors to the displacement of the cable points while preserving the length constraint. This formulation was extended in Almaghout and Klimchik (2022*b*), where a featureless cable was detected and sampled using virtual feature points, and intermediate profiles were generated to guide the cable toward the desired shape in real experiments. Manipulation planning with intermediate cable profiles was developed further in Almaghout and Klimchik (2024), and a later extension addressed large and complex planar deformations by introducing an intermediary-shape generation strategy and an optimization-based co-manipulation controller (Almaghout et al. 2024). These works showed that point-based DLO representations and intermediate shape generation are effective for planar shape control. Their scope, however, is limited to tracking target shapes or profiles that are already given: none of these methods decides how the whole cable should move through an obstacle-filled workspace, which requires planning over full sequences of cable configurations.

Several works address path planning for DLOs using physically motivated models. Early studies formulated DLO planning by sampling stable or minimum-energy configurations (Moll and Kavraki 2006). Bretl and McCarthy (2014) analyzed quasi-static manipulation of Kirchhoff elastic rods and showed that equilibrium configurations can be represented on a finite-dimensional manifold, and Roussel et al. (2020) extended this line of work to elastic rods with contacts. Sintov et al. (2020) reduced online planning time by precomputing a roadmap of stable elastic-rod configurations for dual-arm manipulation. Such rod-based planners represent cable mechanics faithfully, but they depend on equilibrium computation, accurate mechanical parameters, contact-rich simulation, or precomputed roadmaps, which becomes costly when many candidate cable configurations must be generated and checked during a global search.

Recent works have attempted to make DLO planning more deployable in constrained and cluttered environments. Yu et al. (2023) proposed a coarse-to-fine framework for dual-arm manipulation of DLOs with whole-body obstacle avoidance, where a simplified elastic model is used for global planning and a local controller compensates for model errors during execution. Their later work extended this idea to three-dimensional constrained environments by considering the DLO, the robot arms, stable deformation, closed-chain constraints, overstretch prevention, and collision avoidance within a whole-body planning–control framework (Yu et al. 2025). These studies demonstrate that global DLO manipulation in cluttered environments can be achieved when planning is supported by closed-loop feedback, but the global plan serves mainly as a guidance path, and execution robustness depends on the controller compensating for modeling errors and simplifications in the planning model.

A similar planning–control direction was proposed by Aksoy and Wen (2025), whose framework applies an off-the-shelf global planner to the DLO approximated as a chain of rigid links connected by spherical joints. A real-time controller based on control barrier functions and position-based dynamics then enforces obstacle avoidance, stress limits, and DLO integrity. Because no specialized DLO planner has to be designed, the approach can be deployed with standard planning tools; the price is that the global planner produces a coarse path for the approximated object, while fine deformation behavior, stress constraints, and modeling errors are handled mainly by the local controller. In contrast, the objective of GCR is to reason directly about the full planar cable configuration during planning and to validate each accepted planning edge as a local cable deformation.

Other recent approaches reduce the difficulty of constrained DLO planning by using hierarchical representations. Tang et al. (2026) first generate a spatial path set satisfying homotopic constraints for DLO keypoints, then optimize a temporal deformation sequence and track it using a neural model-predictive controller. Reducing the search space before deformation optimization makes such pipelines effective for constrained manipulation, but they do not directly address the specific problem considered here: constructing a compact sequence of full cable configurations in a planar cluttered workspace while explicitly validating each local swept deformation between consecutive configurations.

Cable routing has also been studied in task-specific settings. For example, fixture-based or assembly-oriented methods represent the task through cable features, fixtures, grasp actions, or local routing operations (Jin et al. 2022; Waltersson et al. 2022; Keipour et al. 2022). These methods are efficient when the task structure is known, but they rely on specific fixtures, predefined action primitives, or simplified representations of the cable state, and therefore do not solve the full-shape planning problem where the planner must decide how the entire cable centerline should move through a cluttered environment.

What remains missing across these lines of work is a planner whose search operates on the complete cable shape. Most existing approaches validate static configurations or track planned references, and little attention is given to explicit validation of the swept region generated by the local deformation between two neighboring cable configurations.

The proposed GCR planner addresses this gap for the planar case: every accepted edge is validated as an ordered transition whose swept envelope is collision-free and whose endpoint correspondence is preserved throughout the search. GCR is therefore positioned between lightweight geometric planners and high-fidelity physical DLO planners. It provides a compact global routing layer that reasons about both static cable feasibility and transition-level deformation feasibility.

3. Problem Formulation

Let us represent the deformable linear object (DLO) as an ordered planar polygonal chain. In this case the objective can be formulated as finding a finite sequence of cable configurations that connects a prescribed start configuration to a prescribed goal configuration. In the following, we first describe the workspace and the discrete cable representation. We then introduce the local deformation measures that quantify how the cable changes between two configurations, and use them to define static admissibility, the configuration distance, and the path cost. Finally, we formalize transition feasibility through the swept envelope of a local deformation and extend it to ordered transitions, in which the endpoint correspondence of the cable is taken into account.

3.1. Workspace and Cable Representation

Let

$$\mathscr{W} = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \subset \mathbb{R}^2 \tag{1}$$

be a bounded planar workspace. Where $x_{\min}$ and $x_{\max}$ denote the lower and upper bounds of the workspace along the x-axis, and $y_{\min}$ and $y_{\max}$ denote the corresponding bounds along the y-axis. The obstacle set is denoted by

$$\mathscr{O}_{\text{set}} = \{O_1, O_2, \dots, O_M\}, \tag{2}$$

Where M is the number of static obstacles, and O_m denote the m-th obstacle. Each obstacle is represented either as a polygon or as a circle. The total occupied region is

$$\mathscr{O} = \bigcup_{m=1}^{M} O_m, \tag{3}$$

and the collision-free workspace is

$$\mathscr{F} = \mathscr{W} \setminus \mathscr{O}. \tag{4}$$

The obstacle geometry is assumed to be known and fixed during planning. The implemented benchmark checks collisions against polygonal obstacle boundaries and circular obstacles directly. Therefore, collision validity is evaluated geometrically rather than through an occupancy-grid approximation.

Within this workspace, a cable configuration is represented by an ordered sequence of N planar nodes, where N is the number of cable discretization points. The n-th node is denoted by p_n, with Cartesian coordinates $p_n = (x_n, y_n)$, where x_n and y_n are its coordinates along the x- and y-axes, respectively. Thus,

$$C = (p_1, p_2, \dots, p_N), \qquad p_n = (x_n, y_n) \in \mathbb{R}^2, \qquad n = 1, \dots, N. \tag{5}$$

The cable shape is approximated by the polygonal chain connecting consecutive nodes. The node ordering is part of the configuration: it fixes the endpoint correspondence used for transition validation (Section 3.6). The k-th segment is

$$s_k(C) = [p_k, p_{k+1}], \qquad k = 1, \dots, N-1. \tag{6}$$

The ambient configuration space is therefore

$$\mathbf{X} = \mathbb{R}^{2N}. \tag{7}$$

The start and goal configurations are denoted by

$$C_{\text{start}}, C_{\text{goal}} \in \mathbf{X}. \tag{8}$$

Both configurations are validated before planning. In particular, they must have the same number of nodes, lie inside the workspace, satisfy the prescribed segment-length bounds, and be collision-free as polygonal chains.

The nominal segment length used by the implementation is computed from the start configuration:

$$\ell_0 = \frac{1}{N-1} \sum_{k=1}^{N-1} \|p_{k+1}^{\text{start}} - p_k^{\text{start}}\|. \tag{9}$$

The workspace model ilustrated in Figure 1 is used in the benchmark. The rectangular region defines the bounded workspace $\mathscr{W}$, the red regions represent the static obstacles $\mathscr{O}_{\text{set}}$, and the remaining area corresponds to the collision-free space $\mathscr{F}$.

The same figure also shows the start and goal cable configurations as ordered polygonal chains. The first and last nodes, p_1 and p_N, correspond to the robot end-effectors, while the intermediate nodes discretize the deformable cable geometry.

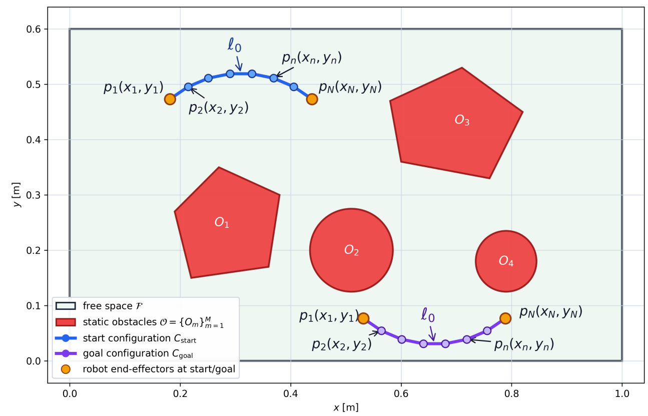


Figure 1: Problem formulation for global cable routing in a planar workspace with static obstacles.

3.2. Local Deformation Measures

The planner itself does not simulate the full dynamic behavior of the cable during search. Instead, it uses local geometric deformation measures to describe how the cable shape changes between two configurations. These measures are based on segment-vector variation and discrete-curvature variation. They are used both to regularize candidate configurations during relaxation and to evaluate the deformation cost of accepted planning edges.

For a cable configuration

$$C = (p_1, p_2, \ldots, p_N), \tag{10}$$

the vector of the k-th cable segment is defined as

$$v_k(C) = p_{k+1} - p_k, \qquad k = 1, \ldots, N-1. \tag{11}$$

This vector represents both the local length and the local orientation of the cable segment. Therefore, changes in $v_k(C)$ describe local stretching, compression, and rotation of the corresponding cable element.

Local bending is described by the second-order finite difference

$$b_k(C) = p_{k+2} - 2p_{k+1} + p_k, \qquad k = 1, \ldots, N-2. \tag{12}$$

This quantity measures the local change of direction along the cable. Larger values correspond to sharper local curvature, while smaller values correspond to smoother local shapes.

Let

$$C^a = (p_1^a, p_2^a, \ldots, p_N^a), \qquad C^b = (p_1^b, p_2^b, \ldots, p_N^b) \tag{13}$$

be two cable configurations with the same number of nodes. The segment-vector change between them is

$$\Delta v_k(C^a, C^b) = v_k(C^a) - v_k(C^b), \qquad k = 1, \ldots, N-1, \tag{14}$$

and the curvature change is

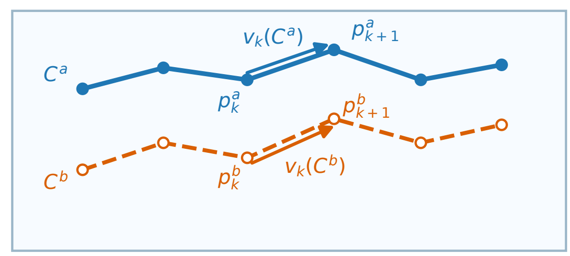
$$\Delta b_k(C^a, C^b) = b_k(C^a) - b_k(C^b), \qquad k = 1, \ldots, N-2. \tag{15}$$

The local deformation energy between the two configurations is then defined as

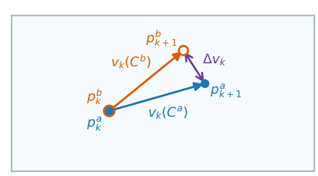
$$E_{\text{def}}(C^a, C^b) = \lambda_s \sum_{k=1}^{N-1} \left\| \Delta v_k(C^a, C^b) \right\|^2 + \lambda_b \sum_{k=1}^{N-2} \left\| \Delta b_k(C^a, C^b) \right\|^2, \tag{16}$$

where $\lambda_s > 0$ and $\lambda_b > 0$ weight the preservation of local segment structure and local curvature, respectively.

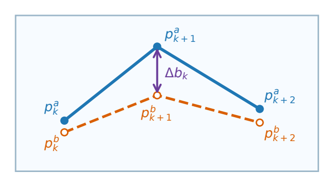
A small value of $E_{\text{def}}(C^a, C^b)$ indicates that the transition preserves the local segment lengths, segment directions, and curvature of the cable. A large value indicates an abrupt local deformation, such as excessive stretching, compression, rotation, or bending. Therefore, E_{def} is used as the deformation penalty in edge-cost computation and path-quality evaluation. The local deformation terms are decomposed in Figure 2 into the segment-vector definition, the corresponding vector change between two neighboring configurations, and the local bending change.



(a) Corresponding segment vectors in C^a and C^b.



(b) Segment-vector change Δv_k.



(c) Local bending change Δb_k.

Figure 2: Local deformation measures between two neighboring cable configurations. Blue solid curves denote C^a, orange dashed curves denote C^b, and the difference arrows indicate the local vector and bending terms that enter the deformation penalty.

The same measure is also used during elastic relaxation. In that case, one of the two configurations is the candidate configuration being optimized, and the other is a reference configuration C^{ref}. This reference-based form reflects the local nature of the planning problem. Minimizing an absolute stretching–bending energy alone could return a smooth, low-energy shape that lies far from the previous feasible configuration and therefore requires an unnecessarily large deformation along a single planning edge. Since each edge in GCR represents a local manipulation step, deformation is instead evaluated relative to a nearby feasible configuration: the planner preserves local cable structure while still allowing the shape to change when obstacle avoidance or tree expansion requires it.

3.3. Admissible and Collision-Free Configurations

A cable configuration is considered statically admissible with respect to a reference configuration when it satisfies three conditions. First, all cable nodes must lie inside the workspace. Second, the deformation of each cable segment must remain bounded with respect to that reference. Third, no cable segment may intersect the obstacle region. As in Section 3.2, the reference is a nearby feasible configuration selected during tree expansion, so the admissibility test is local to the proposed planning edge.

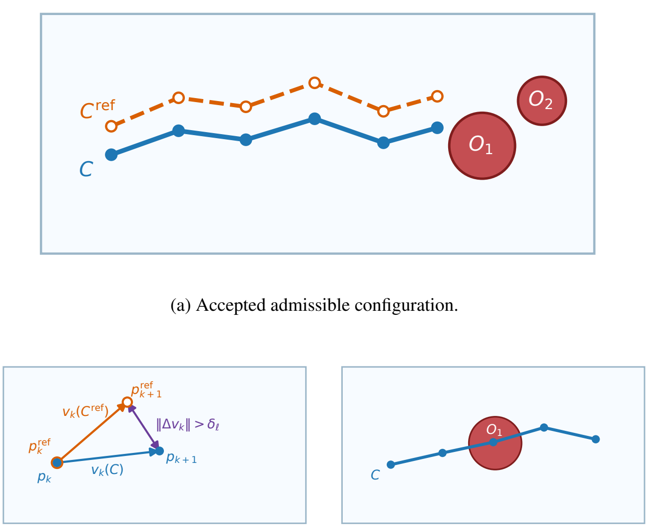
Let C^{ref} denote the feasible reference configuration used to evaluate local deformation. To prevent excessive stretching, compression, or local rotation, the segment-vector deviation $\Delta v_k(C, C^{\text{ref}})$ defined in (14) is bounded as

$$\|\Delta v_k(C, C^{\text{ref}})\| \le \delta_\ell, \qquad k = 1, \dots, N-1, \tag{17}$$

where $\delta_\ell > 0$ is the maximum allowed local segment-vector deviation. Since this bound is evaluated relative to C^{ref}, the admissible set is reference-dependent. We write

$$\begin{aligned} \mathscr{C}_{\text{free}}(C^{\text{ref}}) = \{C \in \mathbf{X} \mid \ & p_n \in \mathscr{W}, && n = 1, \dots, N, \\ & \|\Delta v_k(C, C^{\text{ref}})\| \le \delta_\ell, && k = 1, \dots, N-1, \\ & s_k(C) \cap \mathscr{O} = \emptyset, && k = 1, \dots, N-1\} \end{aligned} \tag{18}$$

and use $\mathscr{C}_{\text{free}}$ as shorthand when the reference configuration is clear from context. Figure 3 illustrates these tests by comparing one accepted candidate with two rejected cases: one violates the local segment-vector bound, and one intersects an obstacle.



(a) Accepted admissible configuration.

(b) Local-bound violation.

(c) Rejected by cable-obstacle intersection.

Figure 3: Schematic examples of static configuration validity. Blue curves denote the candidate configuration, orange dashed curves denote the reference configuration where applicable, and red regions denote obstacles. The rejected cases violate either the local bound $\|\Delta v_k\| \le \delta_\ell$ or the collision-free segment condition $s_k(C) \cap \mathscr{O} = \emptyset$.

This definition checks the complete polygonal representation of the cable, not only its discrete nodes. In particular, obstacle avoidance is verified for every segment

$$s_k(C) = [p_k, p_{k+1}],$$

which prevents collisions along the cable configuration between neighboring nodes.

To measure the safety margin between the cable and the obstacles, we use a signed obstacle distance. For a point $x \in \mathbb{R}^2$, this distance is defined as

$$d_{\mathscr{O}}(x) = \begin{cases} \operatorname{dist}(x, \partial\mathscr{O}), & x \notin \mathscr{O}, \\ 0, & x \in \partial\mathscr{O}, \\ -\operatorname{dist}(x, \partial\mathscr{O}), & x \in \mathscr{O}. \end{cases} \tag{19}$$

Thus, a positive value indicates that the point is outside the obstacle region, a zero value indicates contact with the obstacle boundary, and a negative value indicates penetration into an obstacle.

The clearance of a complete cable configuration is evaluated by sampling points along all cable segments and taking the smallest signed distance among them. For the q-th sample on segment $s_k(C)$, let

$$x_{kq}(C) = (1-\rho_q)p_k + \rho_q p_{k+1}, \qquad \rho_q = \frac{q}{Q}, \qquad q = 0, \dots, Q.$$

The configuration clearance is then

$$\operatorname{clear}(C) = \min_{\substack{k=1,\dots,N-1 \\ q=0,\dots,Q}} d_{\mathscr{O}}\big(x_{kq}(C)\big). \tag{20}$$

In particular, a negative clearance indicates that at least one sampled point lies inside an obstacle and the configuration is in collision.

3.4. Configuration Distance and Path Cost

The distance between two cable configurations is measured by the maximum point-wise displacement of corresponding cable nodes:

$$d(C_i, C_j) = \max_{1 \le n \le N} \left\| p_n^i - p_n^j \right\|. \tag{21}$$

This metric limits the largest motion of any discretized cable point during a local transition. It is therefore suitable for nearest-neighbor search, connection-radius tests, local edge evaluation, and path-cost computation.

A discrete cable path is represented as a finite sequence of configurations,

$$\mathscr{P} = \{C_0, C_1, \dots, C_K\}, \qquad C_0 = C_{\text{start}}, \qquad C_K = C_{\text{goal}}. \tag{22}$$

Each consecutive pair (C_i, C_{i+1}) defines one local deformation of the cable. The geometric length of the path is computed as

$$J_{\text{geom}}(\mathscr{P}) = \sum_{i=0}^{K-1} d(C_i, C_{i+1}). \tag{23}$$

In addition to geometric displacement, the planner penalizes deformation between consecutive cable configurations. The deformation cost of the path is defined as

$$J_{\text{def}}(\mathscr{P}) = \sum_{i=0}^{K-1} E_{\text{def}}(C_{i+1}, C_i), \tag{24}$$

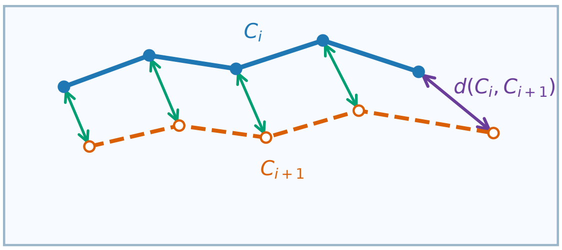
where E_{def} is the local deformation energy defined in (16).

The complete path objective combines the geometric motion cost and the deformation cost:

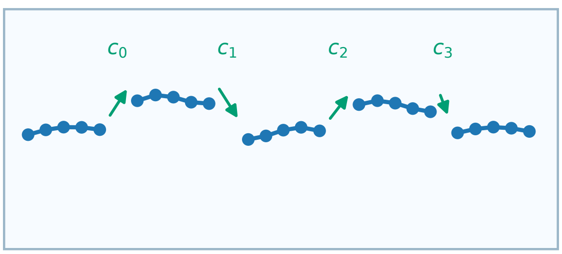
$$J(\mathscr{P}) = J_{\text{geom}}(\mathscr{P}) + \alpha_{\text{def}} J_{\text{def}}(\mathscr{P}), \tag{25}$$

where $\alpha_{\text{def}} \ge 0$ is a weighting coefficient that controls the relative importance of deformation minimization.

The first term favors short transitions in configuration space, while the second term favors smooth and mechanically plausible cable deformation. Figure 4 summarizes the max-norm edge metric and the accumulation of accepted local transition costs along a discrete cable path.



(a) Edge-distance metric.



(b) Accumulated path cost.

Figure 4: Configuration distance and accumulated path cost. The edge-distance panel depicts the max-norm metric $d(C_i, C_{i+1}) = \max_n \|p_n^i - p_n^{i+1}\|$, for which the edge length is induced by the largest displacement among corresponding cable nodes. The path-cost panel shows the accumulation $J(\mathscr{P}) = \sum_i c_i$, where $c_i = d(C_i, C_{i+1}) + \alpha_{\mathrm{def}} E_{\mathrm{def}}(C_{i+1}, C_i)$.

3.5. Transition Feasibility

Static feasibility of individual cable configurations is not sufficient for cable manipulation. Even if two configurations are separately collision-free, the motion between them may pass through an obstacle. Therefore, every edge between two neighboring configurations must be checked for transition feasibility.

Let

$$C^a = (p_1^a, p_2^a, \ldots, p_N^a), \qquad C^b = (p_1^b, p_2^b, \ldots, p_N^b)$$

be two cable configurations. During the transition from C^a to C^b, the cable occupies a swept region in the workspace. The swept region is evaluated with respect to the ordered point correspondence $p_n^a \leftrightarrow p_n^b$. For the k-th segment, a conservative planar swept envelope is the convex hull of the segment before and after the local deformation:

$$\mathscr{E}_{ab}^{(k)} = \mathrm{conv}\left\{p_k^a, p_{k+1}^a, p_{k+1}^b, p_k^b\right\}, \qquad k = 1, \ldots, N-1. \tag{26}$$

The swept envelope of the complete cable transition is then

$$\mathscr{E}_{ab} = \bigcup_{k=1}^{N-1} \mathscr{E}_{ab}^{(k)}. \tag{27}$$

This segment-wise definition avoids relying on a single full-chain polygon when the two cable configurations are folded or when their outer envelope would be self-intersecting. It also makes explicit that the transition depends on the chosen ordering of corresponding nodes.

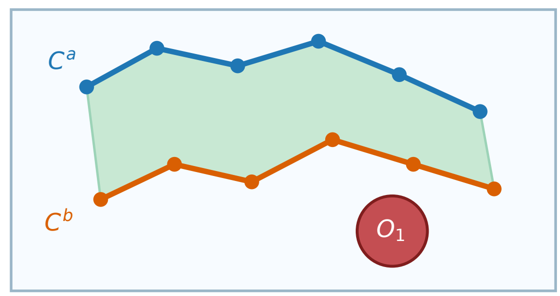
The transition is considered deformation-feasible if this swept envelope does not intersect the obstacle region:

$$\mathscr{E}_{ab} \cap \mathscr{O} = \emptyset. \tag{28}$$

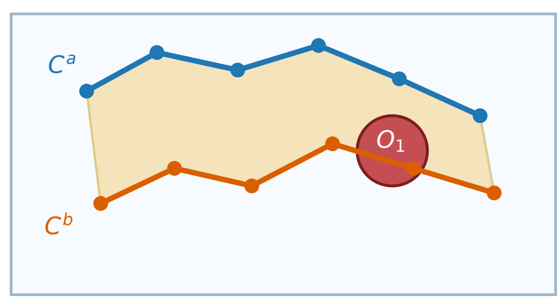
Accordingly, we define the binary swept-clear indicator

$$\chi_{\mathrm{sw}}(C^a, C^b) = \begin{cases} 1, & \mathscr{E}_{ab} \cap \mathscr{O} = \emptyset, \\ 0, & \text{otherwise.} \end{cases} \tag{29}$$

Figure 5 contrasts a clear swept envelope with a blocked one: obstacle avoidance is checked for the continuous local deformation, not only at the two endpoint configurations.



(a) Clear swept envelope.



(b) Blocked swept envelope.

Figure 5: Transition feasibility between two ordered configurations. Blue and orange curves denote the two neighboring cable shapes, the shaded band is the swept envelope, and red regions denote obstacles. The complete edge is accepted only when the swept region is obstacle-free and the remaining conditions in $\Gamma(\cdot,\cdot)$ are also satisfied.

3.6. Ordered Transition Feasibility and Endpoint Correspondence

The swept envelope in (27) is built from the point-to-point correspondence $p_n^a \leftrightarrow p_n^b$, so transition feasibility depends on the assumed node ordering and not only on the geometric shapes of the two configurations. The ordering also determines which cable endpoint is associated with which robot end-effector. This subsection makes the role of the ordering explicit.

For a cable configuration

$$C = (p_1, p_2, \ldots, p_N), \tag{30}$$

we define the reversal operator

$$R(C) = (p_N, p_{N-1}, \ldots, p_1). \tag{31}$$

The configurations C and $R(C)$ describe the same occupied polygonal chain. Therefore, for static collision checking, they are geometrically equivalent. However, they do not necessarily define the same transition when connected to another ordered configuration.

Let

$$C = (p_1, p_2, \ldots, p_N), \qquad D = (q_1, q_2, \ldots, q_N) \tag{32}$$

be two ordered cable configurations. The swept-clear indicator is ordered: $\chi_{\mathrm{sw}}(C, D)$ assumes that point p_n moves toward point q_n for all $n = 1, \ldots, N$. Thus, the transition is defined by the ordered correspondence

$$p_n \longleftrightarrow q_n, \qquad n = 1, \ldots, N. \tag{33}$$

If one of the two configurations is reversed, this correspondence changes. For example, the transition from $R(C)$ to D connects p_{N-n+1} to q_n, which may generate a different swept region from the one generated by the transition from C to D.

This motivates the following observation.

Lemma: Simultaneous Reversal Invariance. Let C and D be two ordered cable configurations. Then

$$\chi_{\mathrm{sw}}(C, D) = \chi_{\mathrm{sw}}(R(C), R(D)). \tag{34}$$

Proof. The transition from C to D connects each point p_n to the corresponding point q_n. If both configurations are reversed, the same geometric correspondence is preserved under the index substitution $n \mapsto N - n + 1$. Therefore, the swept region generated by the ordered pair (C, D) is identical to the swept region generated by the ordered pair $(R(C), R(D))$. Since the obstacle set is unchanged, the collision result is also unchanged. Hence

$$\chi_{\mathrm{sw}}(C, D) = \chi_{\mathrm{sw}}(R(C), R(D)). \tag{35}$$

In contrast, the invariance does not extend to the case where only one configuration is reversed. In general,

$$\chi_{\mathrm{sw}}(C, D) \neq \chi_{\mathrm{sw}}(R(C), D), \tag{36}$$

and

$$\chi_{\mathrm{sw}}(C, D) \neq \chi_{\mathrm{sw}}(C, R(D)). \tag{37}$$

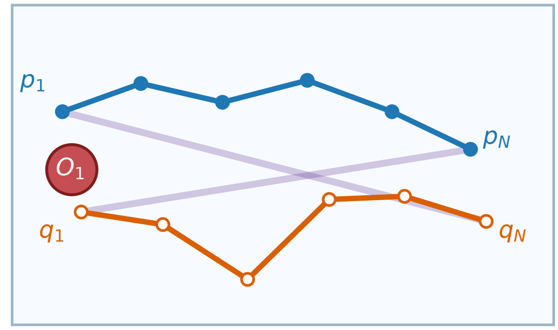
In such cases, the swept envelope may become larger, self-intersecting, or cover a different part of the workspace, so a transition that is collision-free under one ordering may be invalid under another. Figure 6 illustrates this dependence on endpoint correspondence: the same two geometric cable shapes can produce different swept regions when one ordering is changed.

For this reason, transition feasibility must be evaluated for the actual ordered representatives used by the planner. We introduce the endpoint-order variable

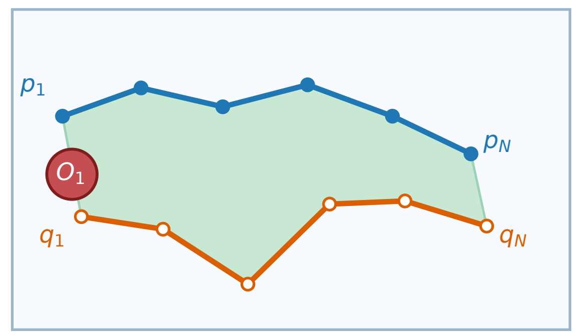
$$\sigma \in \{+1, -1\}, \tag{38}$$

where

$$C^{\sigma} = \begin{cases} C, & \sigma = +1, \\ R(C), & \sigma = -1. \end{cases} \tag{39}$$



(a) Swapped endpoint correspondence.



(b) Preserved endpoint correspondence.

Figure 6: Endpoint correspondence for ordered cable transitions. The same two cable shapes can produce different swept regions under different endpoint correspondences. In this schematic example, the swapped correspondence produces an X-shaped sweep away from the obstacle, whereas the preserved correspondence intersects the obstacle; therefore, χ_{sw} and Γ must be evaluated using the ordered representatives actually used by the planner.

An ordered transition from C^{σ_C} to D^{σ_D} is admissible only if the swept-envelope condition is satisfied for this particular choice of ordering:

$$\chi_{\mathrm{sw}}\left(C^{\sigma_C}, D^{\sigma_D}\right) = 1. \tag{40}$$

In addition to collision-free deformation, the ordered transition should preserve endpoint consistency. We assume that all ordered configurations considered for execution satisfy $p_1 \neq p_N$, so that the endpoint axis is well defined. The ordered endpoint axis of a cable configuration is then defined as

$$u(C) = \frac{p_N - p_1}{\|p_N - p_1\|}. \tag{41}$$

For two ordered configurations C and D, the change in endpoint direction is

$$\theta(C, D) = \arccos\left(u(C)^{\top} u(D)\right). \tag{42}$$

The ordered transition is considered endpoint-consistent if

$$\theta(C, D) \leq \theta_{\max}, \tag{43}$$

where $\theta_{\max}$ is the maximum admissible change in the ordered cable axis.

For compact notation, we define the ordered edge-feasibility predicate by

$$\Gamma(C,D)=1 \iff \begin{cases}\chi_{\mathrm{sw}}(C,D)=1,\\ \theta(C,D)\le\theta_{\max},\\ \Psi(C,D)=1.\end{cases} \tag{44}$$

Here, $\Psi(C,D)$ denotes robot-dependent endpoint feasibility, including admissible endpoint displacement, endpoint obstacle clearance, and end-effector motion limits. If robotic execution constraints are not considered during planning, $\Psi(C,D)$ can be set identically equal to one.

Thus, the ordering used by each accepted tree edge must be fixed during planning, and the ordered transition predicate $\Gamma(\cdot,\cdot)$ must be evaluated before inserting a new edge into the search tree.

4. Global Cable Routing Algorithm

The implemented GCR planner is a bidirectional sampling-based planner in the discretized cable configuration space. It grows one tree from C_{start} and one tree from C_{goal}. Each accepted vertex is a statically valid cable configuration, and each accepted edge satisfies the ordered transition predicate $\Gamma(\cdot,\cdot)$.

4.1. Search Trees and Bidirectional Expansion

Let

$$T_s=(V_s,E_s),\qquad T_g=(V_g,E_g) \tag{45}$$

denote the start and goal trees. They are initialized as

$$V_s=\{C_{\mathrm{start}}\},\qquad V_g=\{C_{\mathrm{goal}}\}. \tag{46}$$

The algorithm alternates between the two trees. At odd iterations, T_s is expanded toward C_{goal}. At even iterations, T_g is expanded toward C_{start}.

For a new accepted configuration C^{new} with parent C^{par}, the stored cost-to-come is

$$g(C^{\mathrm{new}})=g(C^{\mathrm{par}})+d(C^{\mathrm{par}},C^{\mathrm{new}})+\alpha_{\mathrm{def}}E_{\mathrm{def}}(C^{\mathrm{new}},C^{\mathrm{par}}), \tag{47}$$

where $\alpha_{\mathrm{def}}\ge 0$ weights the deformation term. This cost is used as an internal edge-quality measure and is consistent with the path objective $J(\mathscr{P})$ in (25). The corresponding bidirectional structure is shown in Figure 7, where the start and goal trees grow through the obstacle field and are connected only by a validated bridge.

4.2. Guided Configuration Sampling

At each tree-expansion step, the planner generates a raw candidate cable configuration before applying APF biasing and elastic relaxation. Instead of sampling only from a uniform distribution over the configuration space, the proposed method uses a guided mixture sampler. This sampler combines directed sampling toward the target, local expansion from the active tree, and broad exploration of the workspace.

The raw configuration is sampled from the mixture distribution

$$C^{\mathrm{raw}}\sim p_{\mathrm{goal}}\,\mathscr{Q}_{\mathrm{goal}}+p_{\mathrm{tree}}\,\mathscr{Q}_{\mathrm{tree}}+p_{\mathrm{broad}}\,\mathscr{Q}_{\mathrm{broad}}, \tag{48}$$

where $\mathscr{Q}_{\mathrm{goal}}$, $\mathscr{Q}_{\mathrm{tree}}$, and $\mathscr{Q}_{\mathrm{broad}}$ denote the goal-guided, tree-guided, and broad exploration sampling distributions, respectively. Their probabilities satisfy

$$p_{\mathrm{goal}}+p_{\mathrm{tree}}+p_{\mathrm{broad}}=1. \tag{49}$$

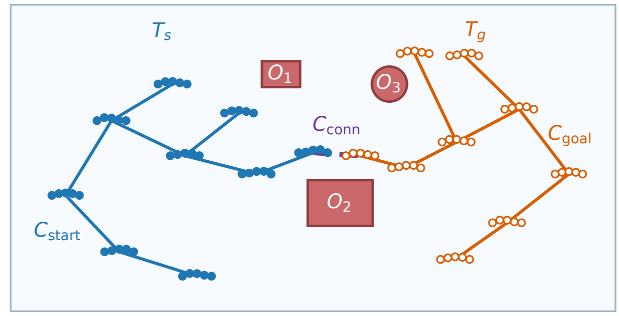


Figure 7: Bidirectional search-tree structure used by GCR. The blue start tree T_s is rooted at C_{start}, and the orange goal tree T_g is rooted at C_{goal}. Red regions are obstacles, and the dashed purple bridge denotes the candidate inter-tree connection C_{conn}. The displayed vertices and edges are collision-free; a bridge is accepted only if the ordered edge-feasibility predicate Γ validates the connection.

The goal-guided component $\mathscr{Q}_{\mathrm{goal}}$ generates candidate configurations in the neighborhood of the current target configuration C^{target}. This component increases the probability of producing samples that move the search toward the opposite tree. A small perturbation is added to avoid repeatedly generating identical configurations and to preserve local exploration around the target shape.

The tree-guided component $\mathscr{Q}_{\mathrm{tree}}$ expands from an existing configuration in the active tree. Let $C^p\in V_{\mathrm{act}}$ be a selected parent configuration and let C^ℓ be a local target configuration. The base configuration is generated by moving from C^p toward C^ℓ:

$$C^{\mathrm{base}}=C^p+\alpha\left(C^\ell-C^p\right), \tag{50}$$

where $\alpha\in[\alpha_{\min},\alpha_{\max}]$ controls the local expansion step. It is chosen according to the distance between C^p and C^ℓ:

$$\alpha=\min\left\{\alpha_{\max},\max\left\{\alpha_{\min},\frac{\gamma\varepsilon}{d(C^p,C^\ell)}\right\}\right\}, \tag{51}$$

where ε is the local connection radius and $\gamma>0$ is a scaling coefficient. This rule keeps the sampled configuration within a controlled neighborhood of the active tree while still directing the expansion toward a useful local target.

After a base configuration has been selected, local perturbations are applied to increase sampling diversity. For each cable node, the sampled point is written as

$$p_i^{\mathrm{raw}}=p_i^{\mathrm{base}}+\xi_i+a\sin\left(\frac{\pi(i-1)}{N-1}\right)n_i(C^{\mathrm{base}}), \tag{52}$$

where ξ_i is a small zero-mean Gaussian perturbation, a is a scalar lateral perturbation amplitude, and $n_i(C^{\mathrm{base}})$ is the local normal direction of the cable at node i. The sinusoidal factor makes the lateral deformation smooth along the cable and reduces perturbations near the endpoints.

The broad exploration component $\mathscr{Q}_{\mathrm{broad}}$ generates a new cable shape using a random-walk construction. Starting from an initial node, successive cable nodes are generated as

$$p_{k+1}=p_k+\ell_0\begin{bmatrix}\cos\theta_k\\ \sin\theta_k\end{bmatrix},\qquad k=1,\dots,N-1, \tag{53}$$

with the heading angle updated according to

$$\theta_{k+1} = \theta_k + \eta_k, \qquad \eta_k \sim \mathcal{N}(0, \sigma_\theta^2). \tag{54}$$

Here, ℓ_0 is the nominal segment length and σ_θ controls the smoothness of the generated shape. After generation, the cable is shifted into the workspace, clipped to the workspace boundaries if necessary, and projected toward the nominal segment length.

The raw sample drawn from this mixture is generally not yet a useful tree candidate; it is first refined by the APF biasing step described next.

4.3. Configuration-Level Artificial Potential Field Biasing

After a raw cable configuration is generated by the guided sampler, it is biased using a configuration-level artificial potential field (APF). The purpose of this step is to move the sampled cable toward a more promising region of the configuration space and push it away from obstacles before the constrained relaxation stage.

Let

$$C^{\text{raw}} = (p_1^{\text{raw}}, p_2^{\text{raw}}, \ldots, p_N^{\text{raw}})$$

be the raw configuration obtained from the guided mixture sampler in (48). Rather than being inserted directly into the search tree, it is first transformed into an APF-biased configuration

$$\widehat{C} = (\widehat{p}_1, \widehat{p}_2, \ldots, \widehat{p}_N)$$

Since a cable configuration is represented as

$$C = (p_1, p_2, \ldots, p_N) \in \mathbb{R}^{2N}, \tag{55}$$

the potential field is defined at the configuration level as a scalar function

$$U : \mathbb{R}^{2N} \to \mathbb{R}. \tag{56}$$

Its gradient is

$$\nabla U(C) = \left(\frac{\partial U}{\partial p_1}, \frac{\partial U}{\partial p_2}, \ldots, \frac{\partial U}{\partial p_N} \right), \tag{57}$$

and the corresponding artificial force applied to the n-th cable point is

$$F_n(C) = -\frac{\partial U}{\partial p_n} \tag{58}$$

The total configuration-level potential is defined as

$$U(C) = U_{\text{att}}(C, C_{\text{target}}) + U_{\text{rep}}(C, \mathcal{O}) \tag{59}$$

Here, C_{target} is the root of the opposite tree. This bidirectional use of the APF is illustrated in Fig. 8: during start-tree expansion the attractive component pulls the sampled configuration toward C_{goal} (Fig. 8a), during goal-tree expansion it pulls toward C_{start} (Fig. 8b), and in both cases obstacle regions generate high potential values that form repulsive zones around occupied areas.

The attractive term guides the cable toward the target configuration:

$$U_{\text{att}}(C, C_{\text{target}}) = \frac{k_{\text{att}}}{2} \sum_{n=1}^{N} \left\| p_n - p_n^{\text{target}} \right\|^2 \tag{60}$$

where $k_{\text{att}} > 0$ is the attractive gain and

$$C_{\text{target}} = \left(p_1^{\text{target}}, p_2^{\text{target}}, \ldots, p_N^{\text{target}} \right) \tag{61}$$

The repulsive term pushes the cable away from obstacles. Since collisions may occur along cable segments and not only at cable nodes, each segment is sampled at $Q+1$ points. For segment $s_k(C)$, the q-th sampled point is

$$s_{kq}(C) = (1-\rho_q)p_k + \rho_q p_{k+1}, \ \rho_q = \frac{q}{Q}, \ q = 0, \ldots, Q \tag{62}$$

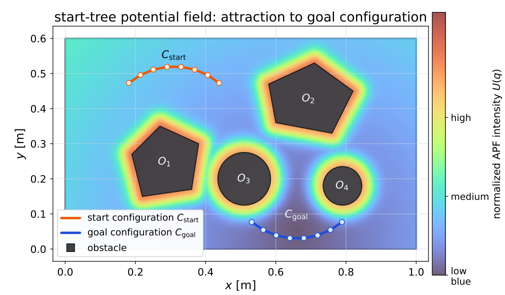
The distance from this point to the obstacle region is

$$r_{kq}(C) = \text{dist}\left(s_{kq}(C), \mathcal{O}\right) = \min_{x \in \mathcal{O}} \left\| s_{kq}(C) - x \right\| \tag{63}$$

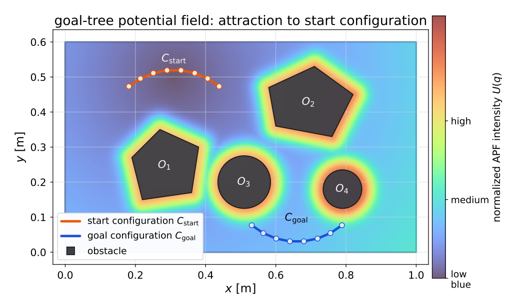
The repulsive potential is then written as

$$U_{\text{rep}}(C, \mathcal{O}) = \sum_{k=1}^{N-1} \sum_{q=0}^{Q} \beta \exp\left(-\frac{r_{kq}^2(C)}{2\sigma^2} \right), \tag{64}$$

where $\beta > 0$ controls the repulsion strength and $\sigma > 0$ controls the obstacle influence range.



(a) Start-tree APF with attraction toward the goal configuration.



(b) Goal-tree APF with attraction toward the start configuration.

Figure 8: Bidirectional configuration-level APF used during tree expansion. The color map represents the normalized potential intensity. Obstacle regions and their neighborhoods correspond to high potential values, while the attractive component biases the expansion toward the opposite tree.

Starting from C^{raw}, the APF-biased sample is generated by H gradient descent steps:

$$C^{(h+1)} = C^{(h)} - \eta \nabla U(C^{(h)}), \qquad h = 0, \ldots, H-1, \tag{65}$$

with

$$C^{(0)} = C^{\text{raw}}, \qquad \widehat{C} = C^{(H)}. \tag{66}$$

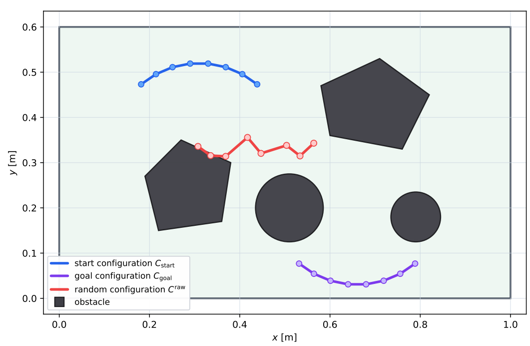
Equivalently, each cable node is updated as

$$p_n^{(h+1)} = p_n^{(h)} + \eta F_n(C^{(h)}), \qquad n = 1, \ldots, N. \tag{67}$$

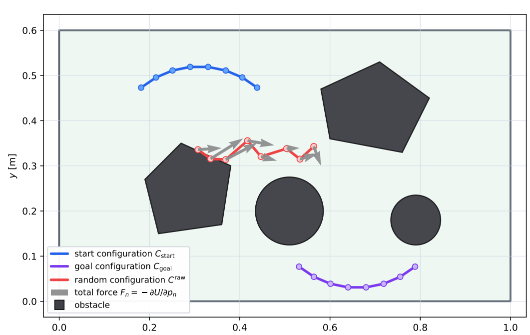
where $\eta > 0$ is the APF step size.

The local effect of this update is shown in Fig. 9. First, a random configuration C^{raw} is generated in the workspace, as shown in Fig. 9a. Then, the gradient of the total potential produces pointwise artificial forces acting on the cable nodes, as shown in Fig. 9b. The raw sample is then transformed into an APF-biased configuration $\widehat{C}$, as shown in Fig. 9c. The biased configuration is closer to the desired expansion direction and is less likely to pass through high-potential obstacle regions.

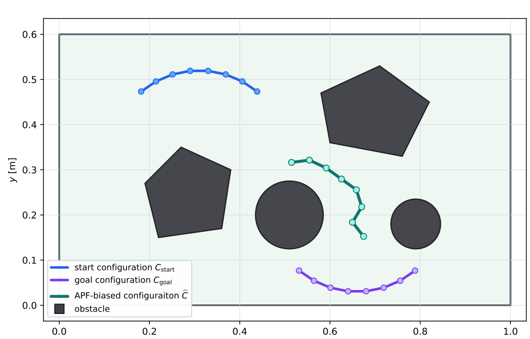
The resulting configuration $\widehat{C}$ is not yet guaranteed to satisfy all cable constraints, such as exact segment-length preservation, admissible deformation, or complete collision avoidance; these violations are handled by the relaxation step described next.



(a) Sampled raw configuration C^{raw}.



(b) Pointwise artificial forces $F_n(C)$ acting on cable nodes.



(c) APF-biased configuration $\widehat{C}$ after gradient descent.

Figure 9: Pipeline of configuration-level APF sampling. A raw random configuration is first generated, then the total APF force is computed for each cable node, and finally the configuration is shifted by gradient descent toward a more promising expansion target.

4.4. Cable Elastic Relaxation

The projected elastic relaxation step transforms the APF-biased configuration $\widehat{C}$ into a corrected candidate

$$C^{\text{relax}} = (p_1^{\text{relax}}, p_2^{\text{relax}}, \dots, p_N^{\text{relax}})$$

that remains close to $\widehat{C}$ while preserving the local shape of a nearby feasible reference configuration C^{ref}.

Let $C = (p_1, p_2, \dots, p_N)$ be the configuration optimized during relaxation. The relaxation objective is defined as

$$\Phi(C) = \lambda_{\text{apf}} \sum_{n=1}^{N} \|p_n - \widehat{p}_n\|^2 + E_{\text{def}}(C, C^{\text{ref}}) \tag{68}$$

The first term keeps the relaxed configuration close to the APF-biased configuration. The deformation term $E_{\text{def}}(C, C^{\text{ref}})$ contains the segment-vector and discrete-curvature penalties defined in (16); these terms reduce excessive stretching, compression, local rotation, and abrupt bending relative to the reference configuration. The weight $\lambda_{\text{apf}} > 0$ controls the relative influence of tracking the APF-biased sample.

The relaxation is initialized from the APF-biased configuration:

$$C^0 = \widehat{C}. \tag{69}$$

At iteration m, a gradient step is first applied:

$$C^{m+\frac{1}{2}} = C^m - h_{\text{relax}} \nabla \Phi(C^m), \tag{70}$$

where $h_{\text{relax}} > 0$ is the relaxation step size.

The intermediate configuration is then corrected by a sequence of projection operators:

$$C^{m+1} = \mathsf{P}_{\text{end}} \mathsf{P}_{\mathscr{W}} \mathsf{P}_{\delta} \mathsf{P}_{\text{obs}} \left(C^{m+\frac{1}{2}} \right). \tag{71}$$

Each projection has a specific role. The obstacle projection P_{obs} pushes cable samples away from obstacles when the required clearance is violated. The segment-deviation projection P_{δ} enforces an upper bound on local deformation with respect to the reference shape. The workspace projection $\mathsf{P}_{\mathscr{W}}$ keeps all cable nodes inside the bounded planning workspace. Finally, the endpoint projection P_{end} restores the APF-updated endpoint positions, so that only the internal cable nodes are modified by the relaxation process.

Obstacle clearance is evaluated at sampled points along each cable segment:

$$x_{kq}(C) = (1 - \rho_q) p_k + \rho_q p_{k+1}, \qquad \rho_q = \frac{q}{Q}, \qquad q = 0, \dots, Q. \tag{72}$$

The clearance condition is

$$d_{\mathscr{O}}\big(x_{kq}(C)\big) \geq d_{\text{safe}}, \tag{73}$$

where $d_{\mathscr{O}}(\cdot)$ is the signed obstacle distance and $d_{\text{safe}} > 0$ is the required safety margin. If a sampled point violates this condition, P_{obs} applies a local correction in the outward obstacle-normal direction. This correction is distributed to the two points of the corresponding cable segment. Thus, the projection improves clearance without globally reshaping the entire cable.

The segment-deviation projection limits the change of each local segment vector relative to the reference configuration. For each segment, the admissible deviation is

$$\left\| (p_{k+1} - p_k) - (p_{k+1}^{\text{ref}} - p_k^{\text{ref}}) \right\| \leq \delta_\ell, \qquad k = 1, \dots, N-1, \tag{74}$$

where $\delta_\ell > 0$ is the maximum allowed segment-vector deviation. If this bound is exceeded, P_{δ} shortens the deviation vector to the radius δ_ℓ and updates the corresponding segment nodes accordingly.

The workspace projection enforces the rectangular workspace bounds

$$\mathscr{W} = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}].$$

For a point $p_n = (x_n, y_n)$, this projection clips the coordinates to the workspace limits:

$$\mathsf{P}_{\mathscr{W}}(p_n) = (\mathrm{clip}(x_n, x_{\min}, x_{\max}), \mathrm{clip}(y_n, y_{\min}, y_{\max})), \tag{75}$$

where

$$\mathrm{clip}(a, a_{\min}, a_{\max}) = \min\{a_{\max}, \max\{a_{\min}, a\}\} \tag{76}$$

The endpoint projection $\mathsf{P}_{\mathrm{end}}$ fixes the two cable endpoints:

$$p_1 = \widehat{p}_1, \qquad p_N = \widehat{p}_N \tag{77}$$

After M_{relax} iterations, the relaxed configuration is defined as

$$C^{\mathrm{relax}} = C^{M_{\mathrm{relax}}}. \tag{78}$$

The resulting candidate C^{relax} is considered for insertion into the search tree only if it passes the final admissibility check with respect to the reference configuration,

$$C^{\mathrm{relax}} \in \mathscr{C}_{\mathrm{free}}(C^{\mathrm{ref}}). \tag{79}$$

This check verifies workspace bounds, segment-deviation limits, and collision freedom of all cable segments.

The effect of the relaxation step is illustrated in Fig. 10.

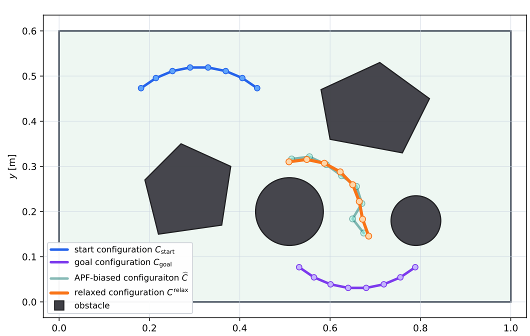


Figure 10: Projected elastic relaxation of an APF-biased cable configuration. The relaxed configuration remains close to the APF-biased sample while improving obstacle clearance and satisfying local segment-deviation constraints.

4.5. Feasible Parent Selection and Node Insertion

After the relaxed configuration C^{relax} has passed the static admissibility check in (79), the planner searches for a feasible parent in the active tree. The parent selection step determines whether the relaxed configuration can be connected to an existing tree vertex by a collision-free and endpoint-consistent local deformation.

In contrast to static configuration validation, local transition validation depends on the ordering of the cable nodes. The planner therefore considers both ordered representatives of the relaxed configuration, $\left(C^{\mathrm{relax}}\right)^\sigma$ with $\sigma \in \{+1, -1\}$, following the endpoint-order notation of Section 3.6. Each vertex already stored in the active tree has a fixed endpoint ordering, because it was inserted through a previously validated ordered transition.

Let V_{act} denote the set of vertices in the active tree. A pair (C_i, σ), where $C_i \in V_{\mathrm{act}}$ and $\sigma \in \{+1, -1\}$, is first considered as a parent-candidate pair only if the corresponding ordered relaxed configuration lies within the local connection radius:

$$d\left(C_i, \left(C^{\mathrm{relax}}\right)^\sigma\right) \leq \varepsilon. \tag{80}$$

This condition restricts the parent search to nearby configurations and avoids large local deformations that may be difficult to validate. The set of nearby parent-candidate pairs is therefore

$$\mathscr{V}_\varepsilon = \left\{(C_i, \sigma) \,\middle|\, C_i \in V_{\mathrm{act}},\ \sigma \in \{+1, -1\},\ d\left(C_i, \left(C^{\mathrm{relax}}\right)^\sigma\right) \leq \varepsilon\right\}. \tag{81}$$

The candidates in $\mathscr{V}_\varepsilon$ are ordered according to their distance from the corresponding ordered relaxed configuration, and only a bounded number of nearest candidates is evaluated.

Distance proximity alone is not sufficient for parent selection. A candidate parent must also produce a deformation-feasible edge under the selected endpoint ordering. For each candidate pair (C_i, σ), let

$$\left(C^{\mathrm{relax}}\right)^\sigma = \left(p_1^{\mathrm{relax},\sigma}, p_2^{\mathrm{relax},\sigma}, \ldots, p_N^{\mathrm{relax},\sigma}\right).$$

The swept envelope between C_i and $\left(C^{\mathrm{relax}}\right)^\sigma$ is constructed as

$$\mathscr{E}^\sigma_{i,\mathrm{relax}} = \bigcup_{k=1}^{N-1} \mathrm{conv}\left\{p_k^i, p_{k+1}^i, p_{k+1}^{\mathrm{relax},\sigma}, p_k^{\mathrm{relax},\sigma}\right\}. \tag{82}$$

The candidate parent is feasible only if this swept envelope does not intersect the obstacle region:

$$\mathscr{E}^\sigma_{i,\mathrm{relax}} \cap \mathscr{O} = \emptyset. \tag{83}$$

Equivalently, the swept-clear condition is written as

$$\chi_{\mathrm{sw}}\left(C_i, \left(C^{\mathrm{relax}}\right)^\sigma\right) = 1. \tag{84}$$

As established in Section 3.6, χ_{sw} is not invariant under reversal of a single configuration, so the parent-selection step evaluates the swept-envelope condition for the actual ordered representative that would be inserted into the tree. To include ordered edge-feasibility consistency in the same edge-validation step, the full ordered transition predicate $\Gamma(\cdot,\cdot)$ of (44) is used. Thus, the feasible parent set is defined as

$$\mathscr{V}_{\mathrm{par}} = \left\{(C_i, \sigma) \in \mathscr{V}_\varepsilon \,\middle|\, \Gamma\left(C_i, \left(C^{\mathrm{relax}}\right)^\sigma\right) = 1\right\}. \tag{85}$$

In particular,

$$\Gamma\left(C_i, \left(C^{\mathrm{relax}}\right)^\sigma\right) = 1 \implies \chi_{\mathrm{sw}}\left(C_i, \left(C^{\mathrm{relax}}\right)^\sigma\right) = 1. \tag{86}$$

If $\mathscr{V}_{\mathrm{par}} = \emptyset$, the relaxed configuration is rejected because no valid ordered local deformation connects it to the active tree.

If at least one feasible parent exists, the parent and the ordering of the relaxed configuration are selected from $\mathscr{V}_{\mathrm{par}}$ by minimizing the resulting cost-to-come. Thus, the selected pair is

$$\begin{aligned}(C^{\mathrm{par}}, \sigma^*) = \arg \min_{(C_i,\sigma)\in\mathscr{V}_{\mathrm{par}}} \Big[& g(C_i) + d\left(C_i, \left(C^{\mathrm{relax}}\right)^\sigma\right) \\ & + \alpha_{\mathrm{def}} E_{\mathrm{def}}\left(\left(C^{\mathrm{relax}}\right)^\sigma, C_i\right)\Big].\end{aligned} \tag{87}$$

The relaxed configuration is then inserted into the active tree using the selected endpoint ordering:

$$C^{\mathrm{new}} = \left(C^{\mathrm{relax}}\right)^{\sigma^*}. \tag{88}$$

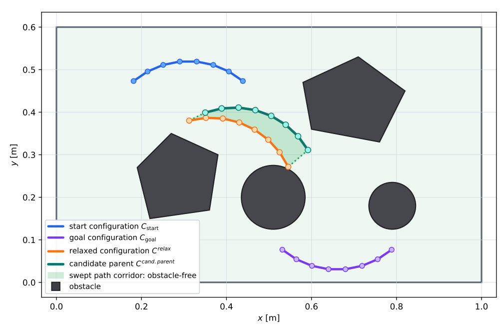
The corresponding edge

$$(C^{\mathrm{par}}, C^{\mathrm{new}}) \tag{89}$$

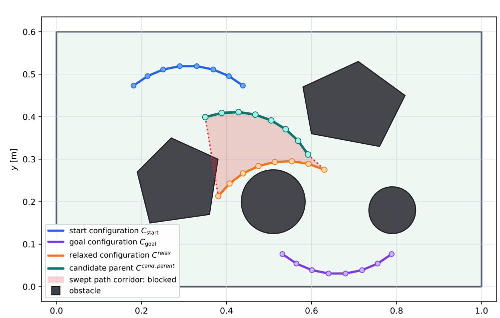
is added to the active tree, and the parent pointer of C^{new} is set to C^{par}.

By construction, every inserted edge satisfies $\Gamma(C^{\mathrm{par}}, C^{\mathrm{new}}) = 1$ and hence, by (86), represents an obstacle-free swept deformation under the selected endpoint correspondence. This prevents a configuration from being accepted under one endpoint ordering and later reinterpreted using another ordering that was not validated during planning.

The two possible outcomes of the parent-selection test are illustrated in Fig. 11. In Fig. 11a, the swept corridor between the candidate parent and the ordered relaxed configuration is obstacle-free. Therefore, the candidate parent is accepted and the ordered relaxed configuration can be inserted into the active tree. In Fig. 11b, the swept corridor intersects an obstacle. Although the two endpoint configurations may be individually valid, the ordered deformation between them is not collision-free, and the candidate parent is rejected.



(a) Feasible parent connection. The swept corridor between the candidate parent $C^{\mathrm{cand.parent}}$ and the ordered relaxed configuration C^{new} is obstacle-free.



(b) Rejected parent connection. The swept corridor intersects an obstacle, so the transition from the candidate parent to the ordered relaxed configuration is not swept-clear.

Figure 11: Feasible parent selection using the swept envelope between a candidate parent configuration and the ordered relaxed configuration. The relaxed configuration can be inserted into the active tree only if at least one nearby parent produces an obstacle-free and endpoint-consistent swept corridor.

4.6. Tree Connection and Path Extraction

After a new node C^{new} has been inserted into the active tree, the algorithm attempts to connect it to the opposite tree. Since each tree vertex is stored with a fixed endpoint ordering, the connection step must evaluate the ordered transition between the two meeting configurations. Therefore, the connection between the two trees is validated using the same ordered edge-feasibility predicate $\Gamma(\cdot,\cdot)$ used in the parent-selection stage.

Let V_{opp} denote the vertex set of the opposite tree. The nearest candidate connection node is first defined as

$$C_{\mathrm{conn}} = \arg\min_{C \in V_{\mathrm{opp}}} d(C, C^{\mathrm{new}}). \tag{90}$$

This nearest-neighbor query provides a candidate connection between the two trees. However, distance proximity alone is not sufficient: the connection edge is traversed in the direction of the final path, from C^{new} to C_{conn} when the active tree is T_{s}, and from C_{conn} to C^{new} when the active tree is T_{g}. The ordered transition in this direction must lie within the connection radius ε and satisfy the ordered edge-feasibility predicate $\Gamma(\cdot,\cdot)$; otherwise, an invalid endpoint correspondence could generate a swept envelope that collides during the connection transition even though both meeting configurations are individually collision-free.

If the nearest candidate C_{conn} does not satisfy the ordered connection condition, the planner may evaluate a bounded set of nearby vertices from V_{opp}. This set is defined as

$$\mathscr{V}_{\mathrm{conn}} = \left\{ C_j \in V_{\mathrm{opp}} \;\middle|\; d(C_j, C^{\mathrm{new}}) \le \varepsilon \right\}. \tag{91}$$

When the active tree is T_{s}, the feasible connection set is

$$\mathscr{V}_{\mathrm{conn}}^{\mathrm{feas}} = \left\{ C_j \in \mathscr{V}_{\mathrm{conn}} \;\middle|\; \Gamma(C^{\mathrm{new}}, C_j) = 1 \right\}. \tag{92}$$

When the active tree is T_{g}, the feasible connection set is

$$\mathscr{V}_{\mathrm{conn}}^{\mathrm{feas}} = \left\{ C_j \in \mathscr{V}_{\mathrm{conn}} \;\middle|\; \Gamma(C_j, C^{\mathrm{new}}) = 1 \right\}. \tag{93}$$

If

$$\mathscr{V}_{\mathrm{conn}}^{\mathrm{feas}} = \emptyset, \tag{94}$$

then no valid tree connection is made during the current iteration. Otherwise, the connection node is selected as the nearest feasible connection candidate:

$$C_{\mathrm{conn}} = \arg\min_{C_j \in \mathscr{V}_{\mathrm{conn}}^{\mathrm{feas}}} d(C_j, C^{\mathrm{new}}). \tag{95}$$

If the active tree is T_{s}, the meeting nodes are assigned as

$$C_{\mathrm{meet}}^{\mathrm{s}} = C^{\mathrm{new}}, \qquad C_{\mathrm{meet}}^{\mathrm{g}} = C_{\mathrm{conn}}. \tag{96}$$

If the active tree is T_{g}, the assignment is

$$C_{\mathrm{meet}}^{\mathrm{s}} = C_{\mathrm{conn}}, \qquad C_{\mathrm{meet}}^{\mathrm{g}} = C^{\mathrm{new}}. \tag{97}$$

The start-side partial path is obtained by tracing parent pointers from $C_{\mathrm{meet}}^{\mathrm{s}}$ to the root C_{start} and reversing the order:

$$\mathscr{P}_{\mathrm{s}} = \mathrm{Reverse}\left(\mathrm{TraceParents}\left(C_{\mathrm{meet}}^{\mathrm{s}} \to C_{\mathrm{start}}\right)\right). \tag{98}$$

The goal-side partial path is obtained by tracing parent pointers from $C_{\mathrm{meet}}^{\mathrm{g}}$ to the root C_{goal}:

$$\mathscr{P}_{\mathrm{g}} = \mathrm{TraceParents}\left(C_{\mathrm{meet}}^{\mathrm{g}} \to C_{\mathrm{goal}}\right). \tag{99}$$

Thus, the complete solution path is

$$\mathscr{P} = \mathscr{P}_{\mathrm{s}} \oplus \mathscr{P}_{\mathrm{g}}, \tag{100}$$

where $\oplus$ denotes path concatenation.

By construction, every configuration in $\mathscr{P}$ belongs to $\mathscr{C}_{\mathrm{free}}$, and every edge between consecutive configurations satisfies the ordered deformation-feasibility condition:

$$\Gamma(C_k, C_{k+1}) = 1, \qquad k = 0, \dots, |\mathscr{P}| - 2. \tag{101}$$

Consequently,

$$\chi_{\mathrm{sw}}(C_k, C_{k+1}) = 1, \qquad k = 0, \ldots, |\mathscr{P}| - 2. \tag{102}$$

4.7. Post-Connection Shortcut Refinement

The first path obtained by connecting the two trees may contain unnecessary intermediate configurations. Therefore, after a feasible solution has been found, a shortcut-based refinement step is applied to reduce the number of cable configurations in the path.

Let the initial connected path be

$$\mathscr{P}^0 = \{C_0, C_1, \ldots, C_K\}, \qquad C_0 = C_{\mathrm{start}}, \qquad C_K = C_{\mathrm{goal}} \tag{103}$$

For any pair of configurations C_i and C_j with $j > i+1$, the algorithm attempts to replace the subsequence

$$(C_i, C_{i+1}, \ldots, C_j) \tag{104}$$

by the direct transition (C_i, C_j). This replacement is accepted if it does not increase the accumulated edge cost,

$$d(C_i, C_j) + \alpha_{\mathrm{def}} E_{\mathrm{def}}(C_j, C_i) \leq \sum_{k=i}^{j-1} \left[d(C_k, C_{k+1}) + \alpha_{\mathrm{def}} E_{\mathrm{def}}(C_{k+1}, C_k) \right], \tag{105}$$

and if the direct transition is deformation-feasible:

$$\Gamma(C_i, C_j) = 1 \tag{106}$$

The shortcutting procedure is repeated until no further improvement can be made, producing the compact path

$$\mathscr{P}^{\mathrm{ref}} = \{C_0, C_1, \ldots, C_M\}, \qquad M \leq K \tag{107}$$

Since every accepted shortcut must satisfy (106), the refined path preserves the collision and transition-feasibility properties enforced by the original path checks.

4.8. Summary of the GCR Planner

The proposed GCR planner is a validity-preserving bidirectional search in the discretized cable configuration space. It grows one tree from the start configuration and one tree from the goal configuration. At each iteration, one tree is selected as the active tree and expanded toward the opposite side. A raw cable configuration is generated by the guided sampler, biased by the configuration-level APF, and corrected by projected elastic relaxation. The relaxed candidate is inserted only if it satisfies static admissibility and if there exists a feasible ordered parent connection.

The main difference between GCR and a standard bidirectional RRT is that node insertion is not based only on distance and static collision checking. In GCR, each accepted vertex must belong to $\mathscr{C}_{\mathrm{free}}$, and each accepted edge must satisfy the ordered edge-feasibility predicate $\Gamma(\cdot,\cdot)$ of (44). Therefore, the extracted path is a sequence of collision-free cable shapes connected by ordered, deformation-feasible transitions.

Algorithm 1 gives the global bidirectional structure of the planner, Algorithm 2 details the expansion step, and Algorithm 3 summarizes shortcut refinement, where redundant intermediate configurations are removed only when the replacement edge remains ordered-feasible and does not increase the edge cost.

The planner terminates successfully when it extracts a finite path from C_{start} to C_{goal} whose configurations and edges satisfy (101). If no valid connection between the two trees is found within the maximum number of iterations $I_{\max}$, the planner reports failure.

Algorithm 1 Global Cable Routing Planner

Input: Start configuration C_{start}, goal configuration C_{goal}, obstacle region $\mathscr{O}$, maximum number of iterations $I_{\max}$, and connection radius ε.
Output: A deformation-feasible path $\mathscr{P} = (C_{\mathrm{start}}, \ldots, C_{\mathrm{goal}})$, or failure if no valid path is found.
Begin:

if $C_{\mathrm{start}} \notin \mathscr{C}_{\mathrm{free}} \vee C_{\mathrm{goal}} \notin \mathscr{C}_{\mathrm{free}}$ **then**
  return failure
end if
$V_s \leftarrow \{C_{\mathrm{start}}\}$, $E_s \leftarrow \emptyset$
$V_g \leftarrow \{C_{\mathrm{goal}}\}$, $E_g \leftarrow \emptyset$
$T_s \leftarrow (V_s, E_s)$, $T_g \leftarrow (V_g, E_g)$
$g(C_{\mathrm{start}}) \leftarrow 0$, $g(C_{\mathrm{goal}}) \leftarrow 0$
for $i = 1, \ldots, I_{\max}$ **do**
  if odd(i) **then**
    $T_{\mathrm{act}} \leftarrow T_s$
    $T_{\mathrm{opp}} \leftarrow T_g$
    $C^{\mathrm{target}} \leftarrow C_{\mathrm{goal}}$
    dir $\leftarrow 1$
  else
    $T_{\mathrm{act}} \leftarrow T_g$
    $T_{\mathrm{opp}} \leftarrow T_s$
    $C^{\mathrm{target}} \leftarrow C_{\mathrm{start}}$
    dir $\leftarrow -1$
  end if
  (success, C^{new}) $\leftarrow$ EXPANDTREE($T_{\mathrm{act}}, C^{\mathrm{target}}, \mathscr{O}, \varepsilon$)
  if success $=$ false **then**
    continue
  end if
  $V_{\mathrm{conn}} \leftarrow$ NEAR($T_{\mathrm{opp}}, C^{\mathrm{new}}, \varepsilon$)
  $V_{\mathrm{conn}}^{\mathrm{feas}} \leftarrow$ FEASIBLECONNECTIONS($V_{\mathrm{conn}}, C^{\mathrm{new}}$, dir)
  if $V_{\mathrm{conn}}^{\mathrm{feas}} = \emptyset$ **then**
    continue
  end if
  $C^{\mathrm{conn}} \leftarrow$ NEAREST($V_{\mathrm{conn}}^{\mathrm{feas}}, C^{\mathrm{new}}$)
  if dir $= 1$ **then**
    $C_{\mathrm{meet}}^{s} \leftarrow C^{\mathrm{new}}$
    $C_{\mathrm{meet}}^{g} \leftarrow C^{\mathrm{conn}}$
  else
    $C_{\mathrm{meet}}^{s} \leftarrow C^{\mathrm{conn}}$
    $C_{\mathrm{meet}}^{g} \leftarrow C^{\mathrm{new}}$
  end if
  $\mathscr{P}_s \leftarrow$ TRACEPARENTS($C_{\mathrm{meet}}^{s}, C_{\mathrm{start}}$)
  $\mathscr{P}_s \leftarrow$ REVERSE($\mathscr{P}_s$)
  $\mathscr{P}_g \leftarrow$ TRACEPARENTS($C_{\mathrm{meet}}^{g}, C_{\mathrm{goal}}$)
  $\mathscr{P} \leftarrow \mathscr{P}_s \oplus \mathscr{P}_g$
  $\mathscr{P} \leftarrow$ SHORTCUTREFINE($\mathscr{P}, \mathscr{O}$)
  return $\mathscr{P}$
end for
return failure

The procedure NEAR returns all vertices of the opposite tree located within the connection radius ε from C^{new}, and FEASIBLECONNECTIONS applies the direction-dependent ordered edge-feasibility condition of (92)–(93) according to dir.

5. Simulation and Validation

We validate the proposed Global Cable Routing (GCR) planner in two stages. First, we evaluate the planner in planar benchmark scenarios with static obstacles. This stage measures success rate, computation time, number of search iterations, and compactness of the resulting configuration sequence. Second, we use a MuJoCo dual-arm manipulation setup in which the robot grippers

Algorithm 2 Tree Expansion

Require: Active tree $T_{\text{act}} = (V_{\text{act}}, E_{\text{act}})$, target configuration C^{target}, obstacle region $\mathscr{O}$, connection radius ε
Ensure: Pair (success, C^{new})
Begin:
Sample a raw configuration C^{raw} using the guided mixture sampler
$\widehat{C} \leftarrow \text{APF}(C^{\text{raw}}, C^{\text{target}}, \mathscr{O})$
$C^{\text{ref}} \leftarrow \arg\min_{C_i \in V_{\text{act}}} d(C_i, \widehat{C})$
$C^{\text{relax}} \leftarrow \text{RELAX}(\widehat{C}, C^{\text{ref}}, \mathscr{O})$
if $C^{\text{relax}} = \emptyset$ **then**
  return (false, $\emptyset$)
end if
if $C^{\text{relax}} \notin \mathscr{C}_{\text{free}}(C^{\text{ref}})$ **then**
  return (false, $\emptyset$)
end if
$V_\varepsilon \leftarrow \left\{(C_i, \sigma) \,\middle|\, C_i \in V_{\text{act}},\ \sigma \in \{+1, -1\},\ d\left(C_i, \left(C^{\text{relax}}\right)^\sigma\right) \leq \varepsilon\right\}$
$V_{\text{par}} \leftarrow \left\{(C_i, \sigma) \in V_\varepsilon \,\middle|\, \Gamma\left(C_i, \left(C^{\text{relax}}\right)^\sigma\right) = 1\right\}$
if $V_{\text{par}} = \emptyset$ **then**
  return (false, $\emptyset$)
end if
$(C^{\text{par}}, \sigma^*) \leftarrow \arg\min_{(C_i,\sigma) \in V_{\text{par}}} \left[g(C_i) + d\left(C_i, \left(C^{\text{relax}}\right)^\sigma\right) + \alpha_{\text{def}} E_{\text{def}}\left(\left(C^{\text{relax}}\right)^\sigma, C_i\right)\right]$
$C^{\text{new}} \leftarrow \left(C^{\text{relax}}\right)^{\sigma^*}$
$V_{\text{act}} \leftarrow V_{\text{act}} \cup \{C^{\text{new}}\}$
$E_{\text{act}} \leftarrow E_{\text{act}} \cup \{(C^{\text{par}}, C^{\text{new}})\}$
$\text{parent}(C^{\text{new}}) \leftarrow C^{\text{par}}$
$g(C^{\text{new}}) \leftarrow g(C^{\text{par}}) + d(C^{\text{par}}, C^{\text{new}}) + \alpha_{\text{def}} E_{\text{def}}(C^{\text{new}}, C^{\text{par}})$
return (true, C^{new})

Algorithm 3 Shortcut Refinement

Require: Initial path $\mathscr{P} = \{C_0, C_1, \ldots, C_K\}$, obstacle region $\mathscr{O}$
Ensure: Refined path $\mathscr{P}^{\text{ref}}$
Begin:
$\mathscr{P}^{\text{ref}} \leftarrow \mathscr{P}$
$i \leftarrow 0$
while $i < |\mathscr{P}^{\text{ref}}| - 2$ **do**
  $j \leftarrow |\mathscr{P}^{\text{ref}}| - 1$
  while $j > i + 1$ **do**
    $C_i \leftarrow \mathscr{P}^{\text{ref}}[i]$
    $C_j \leftarrow \mathscr{P}^{\text{ref}}[j]$
    $J_{\text{old}} \leftarrow \sum_{k=i}^{j-1} [d(C_k, C_{k+1}) + \alpha_{\text{def}} E_{\text{def}}(C_{k+1}, C_k)]$
    $J_{\text{new}} \leftarrow d(C_i, C_j) + \alpha_{\text{def}} E_{\text{def}}(C_j, C_i)$
    if $\Gamma(C_i, C_j) = 1$ **and** $J_{\text{new}} \leq J_{\text{old}}$ **then**
      Remove configurations $C_{i+1}, \ldots, C_{j-1}$ from $\mathscr{P}^{\text{ref}}$
      break
    end if
    $j \leftarrow j - 1$
  end while
  $i \leftarrow i + 1$
end while
return $\mathscr{P}^{\text{ref}}$

are rigidly attached to the cable endpoints. This validation checks whether endpoint trajectories generated from the planned cable configurations can reproduce the target cable shape in simulation.

5.1. Benchmark Evaluation

We use the benchmark study to evaluate GCR at the cable-configuration level. Each benchmark scenario contains a bounded planar workspace, a set of static obstacles, an initial cable configuration C_{start}, and a target cable configuration C_{goal}. For each scenario, we run 1000 independent trials with different random seeds. We count a trial as successful when the planner finds a sequence of cable configurations from C_{start} to C_{goal} before reaching the maximum number of iterations. The accepted sequence must satisfy the static configuration validity test, the swept-transition feasibility test, and the ordered edge-feasibility validation described in Section 4.

All benchmark experiments are executed using the single-threaded C++ implementation of GCR. The tests are performed on a laptop equipped with an AMD Ryzen 5 5600H CPU with a 3.30GHz base clock and 24GB of DDR4 memory at 3200MT/s. The reported computation time includes tree expansion, shortcut refinement, and ordered edge-feasibility validation. No GPU acceleration or parallel execution is used in the reported benchmark results.

The benchmark contains ten representative routing scenarios, each named after its obstacle layout: Disc Rows, Cluttered Field, Narrow Gate, Bug Trap, Double Gate, Scattered Discs, Block Turn, L-Corridor, Weave, and Narrow L-Corridor. The scenarios differ in obstacle arrangement, free-space connectivity, passage width, and the amount of deformation required to move the cable from C_{start} to C_{goal}. Some scenarios provide a relatively direct route between the two configurations. Others require the cable to pass through narrow passages, rotate around obstacle boundaries, or follow a less direct deformation sequence. One representative solution for each scenario is shown in Figure 12.

The difficulty of a routing problem depends less on the number of obstacles than on how the obstacles restrict feasible cable transitions, which these solution plots make visible.

We report the following metrics for each benchmark scenario:

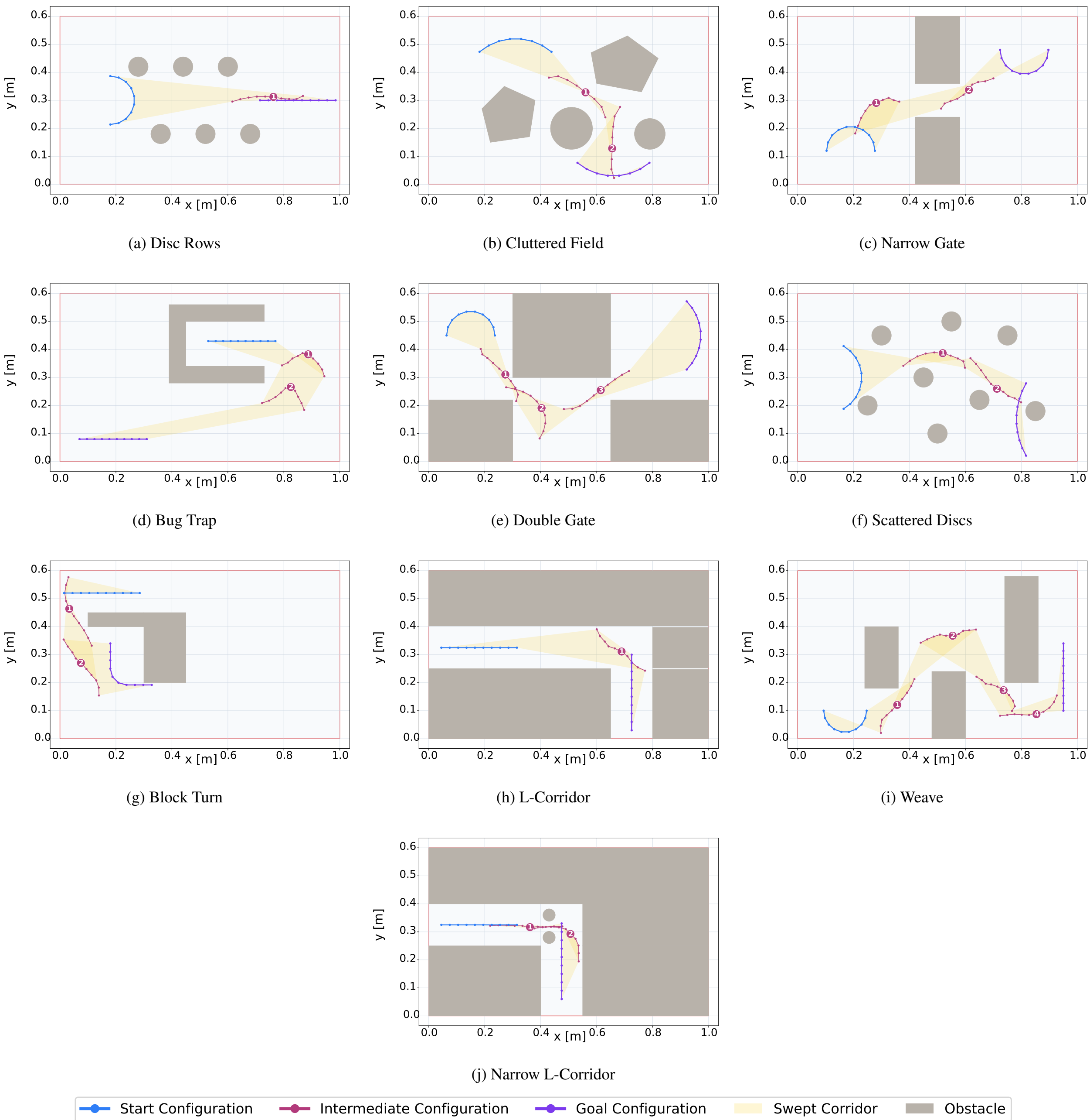
- **Success rate**: the percentage of trials in which GCR finds an accepted path before reaching the maximum number of iterations.
- **Average planning time**: the mean runtime of the complete planning pipeline over successful trials.
- **Fastest planning time**: the minimum runtime observed among the successful trials. This value indicates the best-case behavior of the stochastic search for a given scenario.
- **Average number of iterations**: the mean number of tree-expansion iterations required to obtain a valid connection between the two trees.
- **Configurations before refinement**: the average number of cable configurations in the raw path immediately after the two trees are connected.
- **Configurations after refinement**: the average number of cable configurations remaining after shortcut refinement. This metric describes the compactness of the final path that is later used for execution.

These metrics are used to compare the robustness, computational efficiency, and path compactness of GCR across the benchmark scenarios.

5.2. Benchmark Results

The benchmark results are summarized in Table 1.

GCR found a valid path in every trial, giving a 100% success rate in all ten scenarios. This result should be interpreted within the tested scenario set; it shows that the implemented combination of



(a) Disc Rows (b) Cluttered Field (c) Narrow Gate (d) Bug Trap (e) Double Gate (f) Scattered Discs (g) Block Turn (h) L-Corridor (i) Weave (j) Narrow L-Corridor

Figure 12: Representative GCR solutions for the ten benchmark scenarios. Each subfigure shows the initial cable configuration, the target cable configuration, the static obstacles, and the resulting deformation sequence. Numbered markers denote intermediate cable profiles in path order

bidirectional expansion, APF-biased sampling, elastic relaxation, and ordered swept-transition validation was sufficient for these planar layouts.

The computational cost varies with the geometric constraints of the workspace. **Narrow Gate** (Figure 12c), is solved with the lowest average computation time, 0.008 ± 0.003 s, and requires only $(0.07 \pm 0.03) \times 10^3$ iterations on average. This scenario represents a relatively simple routing case, where the free space allows a direct deformation sequence between C_{start} and C_{goal}. **Double Gate** (Figure 12e), also remains computationally light, with an average time of 0.017 ± 0.005 s and $(0.09 \pm 0.03) \times 10^3$ iterations. Most trials are solved quickly, as indicated by the fastest time of 0.005 s.

The most demanding case is the **Bug Trap** (Figure 12d), where the planner requires 1.306 ± 0.871 s and $(6.98 \pm 4.52) \times 10^3$ iterations on average. This increase is consistent with the more constrained obstacle arrangement, where the cable has fewer admissible deformation routes and the planner must reject more samples because of static collisions or invalid swept transitions. The **Narrow L-Corridor** (Figure 12j), is the second most expensive case, with 0.339 ± 0.139 s and $(2.60 \pm 1.08) \times 10^3$ iterations.

For the remaining scenarios (**Disc Rows**, **Cluttered Field**, **Scattered Discs**, **Block Turn**, **L-Corridor**, and **Weave**), shown in Figure 12, the average runtime ranges from 0.072 s to 0.283 s.

Table 1. : Benchmark performance for the selected GCR scenarios.

Scenario	Success (%)	Time (s)	Fastest (s)	Iterations (10^3)	Raw Path Configurations	Refined Path Configurations
Disc Rows	100.00	0.240 ± 0.143	0.010	2.08 ± 1.20	17 ± 4	5 ± 1
Cluttered Field	100.00	0.283 ± 0.197	0.018	1.75 ± 1.21	20 ± 4	5 ± 1
Narrow Gate	100.00	0.008 ± 0.003	0.002	0.07 ± 0.03	13 ± 2	3 ± 0
Bug Trap	100.00	1.306 ± 0.871	0.124	6.98 ± 4.52	22 ± 5	5 ± 1
Double Gate	100.00	0.017 ± 0.005	0.005	0.09 ± 0.03	16 ± 3	7 ± 0
Scattered Discs	100.00	0.145 ± 0.061	0.014	0.95 ± 0.42	18 ± 4	7 ± 1
Block Turn	100.00	0.151 ± 0.066	0.020	1.36 ± 0.57	19 ± 4	7 ± 1
L-Corridor	100.00	0.072 ± 0.033	0.008	0.65 ± 0.31	16 ± 3	6 ± 1
Weave	100.00	0.104 ± 0.059	0.004	0.72 ± 0.41	18 ± 3	5 ± 1
Narrow L-Corridor	100.00	0.339 ± 0.139	0.054	2.60 ± 1.08	25 ± 4	9 ± 1

These values show that the planner handled the selected static-obstacle tasks efficiently, except when the geometry imposed narrow or highly constrained deformation routes.

The shortcut refinement step also affects the final path representation. Before refinement, the raw connected paths contain between 13 ± 2 and 25 ± 4 configurations on average. After refinement, the final paths contain between 3 and 9 ± 1 configurations. The retained configurations are those that preserve the validity of the cable transition according to the swept-envelope test and ordered edge-feasibility validation.

Overall, the average computation time stays below 0.34 s in nine out of ten scenarios, and the most constrained scenario is solved in about 1.31 s on average. Runtimes of this order leave a comfortable margin for computing cable-routing sequences before execution.

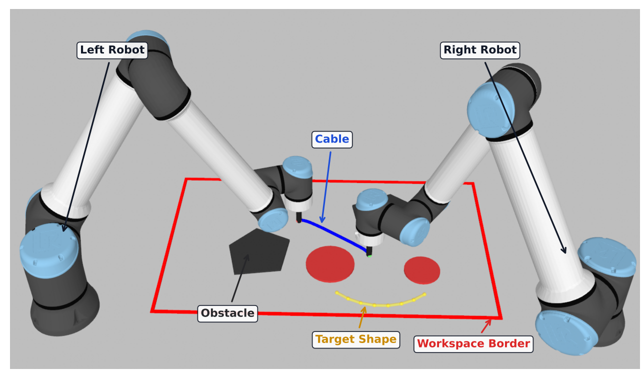


Figure 13: Dual-arm MuJoCo setup used for simulation-based manipulation validation.

5.3. Robotic Manipulation Validation

The second validation stage evaluates whether the refined GCR paths can guide a simulated dual-arm robotic system to deform and translate a cable. The robotic setup, built in MuJoCo (Todorov et al. 2012), is shown in Fig. 13. It contains two UR10 manipulators attached rigidly to the two cable ends, the target cable shape, and the bounded planar workspace with obstacles. The robots track the planned endpoint trajectories, while the internal cable elements move passively according to the simulated cable dynamics, elasticity, gravity, contact response, and endpoint motion.

For each waypoint configuration, the first and last cable points define the Cartesian references for the left and right robot end-effectors. The endpoint motion between two consecutive configurations is executed using smooth task-space interpolation. The settled cable configuration at the end of the execution is then compared with the planned goal configuration.

5.4. Robotic Execution Results

The execution accuracy for the scenarios is reported in Table 2. The root mean square error (RMSE) and maximum point-wise error ($e_{\max}$) are computed over all cable nodes between the settled simulation state and the goal configuration.

$$e_{\max} = \max(e_1, e_2, \dots, e_N) \tag{108}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{n=1}^{N} e_n^2} \tag{109}$$

where e_n is the Euclidean error between the n-th point of the final settled configuration and the corresponding point of the target configuration.

The simulated executions produce final RMSE configuration errors between 1.60 mm and 4.66 mm. For the considered cable length $L = 0.27$ m, this corresponds to approximately 0.59–1.73% of the cable length. The maximum point-wise error ranges from 2.31 mm to 7.81 mm, or approximately 0.86–2.89% of the cable length. These values indicate that the executed cable shape remains close to the planned goal in the tested quasi-static replay conditions.

The error values also reflect the geometry of the tasks. The **Bug Trap** scenario has the lowest RMSE, 1.60 mm, which is consistent with a smoother deformation and relatively simple endpoint motion. **Block Turn** and **Weave** have higher RMSE, 4.66 mm and 4.58 mm, respectively. These cases require stronger deformation and larger cumulative endpoint motion, so small effects from cable

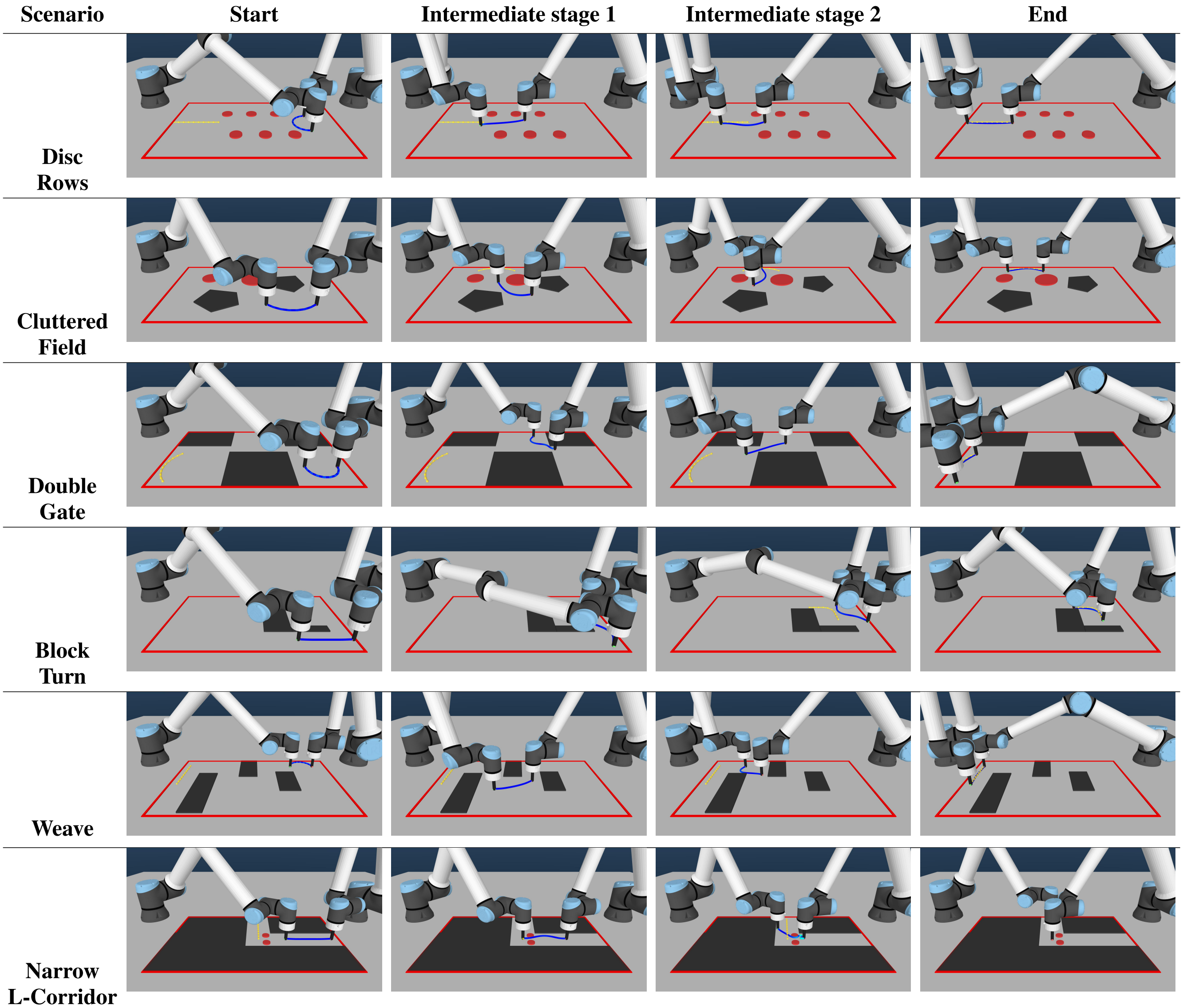


Figure 14: Execution snapshots for the validation scenarios. Each row shows the simulated cable manipulation at the start, two intermediate, and the end configurations.

elasticity, contact response, and numerical settling have a larger influence on the final configuration. Representative executions are shown in Figure 14 at the start, two intermediate stages, and the final state. The manipulation-validation records are available at https://github.com/KaramAlmaghout/gcr.

6. Discussion

The results show that the proposed GCR planner can generate feasible routing paths for a cable-like deformable linear object in two-dimensional cluttered environments. The main contribution of the method is that it plans in the space of complete cable configurations rather than only in the space of endpoint positions. This allows the planner to reason about the shape of the whole cable, including local segment deformation, bending, static collision avoidance, and transition feasibility between consecutive configurations.

A central aspect of the formulation is that every accepted edge, and not only every vertex, is validated: because each neighboring pair of configurations is checked as a local deformation through the swept-envelope condition, the resulting path is more suitable for execution than a sequence of independently valid cable shapes.

The ordered edge-feasibility condition additionally guarantees that the endpoint correspondence used by the planner is preserved in the final path. This matters for dual-arm manipulation, where each cable endpoint is assigned to a specific end-effector and swapping the correspondence would change the executed motion.

The combination of APF-biased sampling and bidirectional tree search provides a practical balance between exploration and guidance. The APF component improves the quality of sampled configurations by moving them toward the opposite tree and away from obstacle regions. At the same time, the planner does not rely on APF descent alone; the bidirectional tree structure preserves global exploration and allows the search to recover from local geometric restrictions.

Table 2. : Robotic manipulation validation results.

Scenario	RMSE (mm)	$e_{\max}$ (mm)
Disc Rows	2.20	3.41
Cluttered Field	2.22	3.72
Narrow Gate	4.32	6.49
Bug Trap	1.60	2.31
Double Gate	1.87	2.70
Scattered Discs	2.08	3.32
Block Turn	4.66	7.81
L-Corridor	4.13	6.28
Weave	4.58	7.32
Narrow L-Corridor	2.83	4.73

The projected elastic relaxation step contributes to the mechanical plausibility of the generated configurations. It corrects APF-biased samples by preserving local segment structure and discrete curvature while keeping the candidate close to the sampled direction. This relaxation is not intended to replace a full physical cable simulator. Instead, it provides a computationally efficient correction step that makes sampled configurations more consistent with the discrete cable model before they are validated and inserted into the tree.

In the benchmark, more constrained scenarios required more iterations and longer planning time, which is expected because narrow passages and obstacle-dense regions reduce the number of admissible local transitions. Shortcut refinement also reduces unnecessary manipulation effort: redundant intermediate configurations are removed only when the direct transition remains feasible, and the deformation cost penalizes abrupt changes in segment vectors and discrete curvature. The final path therefore avoids unnecessary cable reshaping and provides a short sequence of deformation-feasible manipulation steps.

In the manipulation validation, the simulated robots followed the endpoint references and reproduced the planned target cable shapes with final errors below 5 mm RMSE, indicating that the planned sequences are executable by endpoint tracking alone under quasi-static conditions.

The present formulation defines a focused scope for the first implementation of GCR and also indicates several directions for further development. In this work, the planner is evaluated for two-dimensional planar routing with known static obstacles. This setting is relevant to quasi-planar manipulation tasks, but future extensions should consider three-dimensional cable routing, out-of-plane motion, torsion, and contact interactions. The current cable model is geometric and quasi-static: segment and curvature regularization are used to preserve locally plausible shapes, while forces, stress limits, frictional contacts, and dynamic effects are not explicitly simulated during planning. Incorporating such physical quantities would make the method applicable to a wider range of deformable-object manipulation tasks.

The swept-envelope test used in GCR provides an efficient geometric criterion for rejecting unsafe local transitions. Since it approximates the occupied region during a deformation, future work should study more accurate transition models, especially when the real cable motion differs from the assumed interpolation. The validation presented in this paper is limited to a finite set of planar benchmark scenarios and a simulation-based dual-arm setup. These experiments demonstrate the feasibility of the proposed planning pipeline within the tested conditions, while further evaluation is needed across more diverse obstacle layouts, cable stiffness values, robot platforms, and real experimental setups.

The proposed planner is most directly applicable to tasks where a flexible linear object must be routed through a cluttered but mainly planar workspace. Examples include robotic cable routing in electrical cabinet assembly, wire harness installation, routing of flexible tubes or hoses in manufacturing, and planning intermediate shapes for dual-arm manipulation of ropes, wires, or similar objects. In such applications, GCR can serve as a supervisory planner that produces a sequence of feasible cable configurations, while a lower-level controller tracks the corresponding endpoint motions and compensates for execution errors.

7. Conclusion

This paper presented GCR, a global planner for routing cable-like deformable linear objects in two-dimensional cluttered environments. The method represents the cable as an ordered polygonal chain and searches directly in the corresponding configuration space. This allows the planner to evaluate the feasibility of individual cable shapes as well as the feasibility of the local deformations between consecutive configurations.

The proposed planning pipeline combines bidirectional tree search, APF-guided configuration sampling, projected elastic relaxation, ordered transition validation, and shortcut refinement. The benchmark results showed that the planner generated feasible paths in the tested planar scenarios, while the refinement stage reduced redundant intermediate configurations and avoided unnecessary deformation. The simulation-based manipulation validation further indicated that the resulting paths can be converted into endpoint references that reproduce the target cable shapes with small final errors under the tested conditions.

Future work will focus on extending the formulation to three-dimensional environments, incorporating richer physical models of cable deformation and contact, and evaluating the planner on real robotic systems.

References

Aksoy, B. and Wen, J. (2025). Planning and control for deformable linear object manipulation.

Almaghout, K., Cherubini, A., and Klimchik, A. (2024). Robotic co-manipulation of deformable linear objects for large deformation tasks. Preprint.

Almaghout, K. and Klimchik, A. (2022a). Planar shape control of deformable linear objects. *IFAC-PapersOnLine*, 55:2469–2474.

Almaghout, K. and Klimchik, A. (2022b). Vision-based robotic comanipulation for deforming cables. *Russian Journal of Nonlinear Dynamics*, 18(5):843–858.

Almaghout, K. and Klimchik, A. (2024). Manipulation planning for cable shape control. *Robotics*, 13(1):18.

Bretl, T. and McCarthy, Z. (2014). Quasi-static manipulation of a kirchhoff elastic rod based on a geometric analysis of equilibrium configurations. *The International Journal of Robotics Research*, 33(1):48–68.

Caporali, A. and Palli, G. (2026). Robotic perception and manipulation of deformable linear objects: A survey. *The International Journal of Robotics Research*.

Jin, S., Lian, W., Wang, C., Tomizuka, M., and Schaal, S. (2022). Robotic cable routing with spatial representation. *IEEE Robotics and Automation Letters*, 7(2):5687–5694.

Jin, S., Wang, C., and Tomizuka, M. (2019). Robust deformation model approximation for robotic cable manipulation. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 6586–6593.

Karaman, S. and Frazzoli, E. (2011). Sampling-based algorithms for optimal motion planning. *The International Journal of Robotics Research*, 30(7):846–894.

Keipour, A., Bandari, M., and Schaal, S. (2022). Deformable one-dimensional object detection for routing and manipulation. *IEEE Robotics and Automation Letters*, 7(2):4329–4336.

Kuffner, J. J. and LaValle, S. M. (2000). RRT-Connect: An efficient approach to single-query path planning. In *Proceedings of the IEEE International Conference on Robotics and Automation*, pages 995–1001.

Lagneau, R., Krupa, A., and Marchal, M. (2020). Automatic shape control of deformable wires based on model-free visual servoing. *IEEE Robotics and Automation Letters*, 5(4):5252–5259.

Lv, N., Liu, J., and Jia, Y.-B. (2021). Coordinated control of flexible cables with human-like dual manipulators. *Journal of Dynamic Systems, Measurement, and Control*, 143(8):081006.

Moll, M. and Kavraki, L. E. (2006). Path planning for deformable linear objects. *IEEE Transactions on Robotics*, 22(4):625–636.

Navarro-Alarcon, D., Liu, Y.-H., Romero, J. G., and Li, P. (2013). Model-free visually servoed deformation control of elastic objects by robot manipulators. *IEEE Transactions on Robotics*, 29(6):1457–1468.

Roussel, O., Fernbach, P., and Taïx, M. (2020). Motion planning for an elastic rod using contacts. *IEEE Transactions on Automation Science and Engineering*, 17(2):670–683.

Sanchez, J., Corrales, J. A., Bouzgarrou, B.-C., and Mezouar, Y. (2018). Robotic manipulation and sensing of deformable objects in domestic and industrial applications: A survey. *The International Journal of Robotics Research*, 37(7):688–716.

Sintov, A., Macenski, S., Borum, A., and Bretl, T. (2020). Motion planning for dual-arm manipulation of elastic rods. *IEEE Robotics and Automation Letters*, 5(4):6065–6072.

Tang, Y., Yang, T., Huang, J., Chu, X., and Au, K. W. S. (2026). Hierarchical deformation planning and neural tracking for dlos in constrained environments. *IEEE/ASME Transactions on Mechatronics*.

Todorov, E., Erez, T., and Tassa, Y. (2012). Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 5026–5033. IEEE.

Waltersson, G. A., Laezza, R., and Karayiannidis, Y. (2022). Planning and control for cable-routing with dual-arm robot. In *2022 International Conference on Robotics and Automation (ICRA)*, pages 1046–1052. IEEE.

Yin, H., Varava, A., and Kragic, D. (2021). Modeling, learning, perception, and control methods for deformable object manipulation. *Science Robotics*, 6(54):eabd8803.

Yu, M., Lv, K., Wang, C., Jiang, Y., Tomizuka, M., and Li, X. (2025). Generalizable whole-body global manipulation of deformable linear objects by dual-arm robot in 3-d constrained environments. *The International Journal of Robotics Research*, 44(4):607–639.

Yu, M., Lv, K., Wang, C., Tomizuka, M., and Li, X. (2023). A coarse-to-fine framework for dual-arm manipulation of deformable linear objects with whole-body obstacle avoidance. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 10153–10159. IEEE.

Zhu, J., Cherubini, A., Dune, C., Navarro-Alarcon, D., Alambeigi, F., Berenson, D., Ficuciello, F., Harada, K., Kober, J., Li, X., Pan, J., Yuan, W., and Gienger, M. (2022). Challenges and outlook in robotic manipulation of deformable objects. *IEEE Robotics & Automation Magazine*, 29(3):67–77.